

EXERCISES

LEVEL I

(Problems Based on Fundamentals)

SINE RULE

1. If in a triangle ABC , $\frac{a^2 - b^2}{a^2 + b^2} = \frac{\sin(A - B)}{\sin(A + B)}$, then the triangle is right angled or isosceles?
2. Prove that $\frac{\sin(B - C)}{\sin(B + C)} = \frac{b^2 - c^2}{a^2}$
3. In a $\triangle ABC$ such that $\angle A = 45^\circ$ and $\angle B = 75^\circ$ then find $a + c\sqrt{2}$.
4. Prove that
$$\frac{a^2 \sin(B - C)}{\sin B + \sin C} + \frac{b^2 \sin(C - A)}{\sin C + \sin A} + \frac{c^2 \sin(A - B)}{\sin A + \sin B} = 0$$
5. Prove that
$$\frac{b^2 - c^2}{\cos B + \cos C} + \frac{c^2 - a^2}{\cos C + \cos A} + \frac{a^2 - b^2}{\cos A + \cos B} = 0$$
6. In a triangle ABC , if a^2, b^2, c^2 are in AP then prove that $\cot A, \cot B, \cot C$ are in AP.
7. If $\cot \frac{A}{2} = \frac{b + c}{a}$, then prove that $\triangle ABC$ is right angled.
8. Prove that a^2, b^2, c^2 are in A.P., if $\frac{\sin A}{\sin C} = \frac{\sin(A - B)}{\sin(B - C)}$
9. In any $\triangle ABC$, prove that
$$\prod \left(\frac{\sin^2 A + \sin A + 1}{\sin A} \right) > 27.$$
10. In a triangle ABC , prove that,
$$a \sin \left(\frac{A}{2} + B \right) = (b + c) \sin \left(\frac{A}{2} \right)$$
11. In a triangle ABC , prove that,
$$\frac{a \sin(B - C)}{b^2 - c^2} = \frac{b \sin(C - A)}{c^2 - a^2} = \frac{c \sin(A - B)}{a^2 - b^2}$$
12. Prove that $\frac{1 + \cos(A - B) \cos C}{1 + \cos(A - C) \cos B} = \frac{a^2 + b^2}{a^2 + c^2}$
13. In a $\triangle ABC$, if $\cos A + 2 \cos B + \cos C = 2$ then prove that the sides of a triangle are in AP
14. In a $\triangle ABC$, if $\cos A \cos B + \sin A \sin B \sin C = 1$, then prove that $a:b:c = 1:1:\sqrt{2}$.

COSINE RULE

15. In a $\triangle ABC$, prove that
$$a(b \cos C - c \cos B) = b^2 - c^2$$

16. Prove that

$$\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a^2 + b^2 + c^2}{2abc}$$

17. Prove that

$$(a - b)^2 \cos^2 \left(\frac{C}{2} \right) + (a + b)^2 \sin^2 \left(\frac{C}{2} \right) = c^2$$

18. In a $\triangle ABC$, if

$$(a + b + c)(a - b + c) = 3ac, \text{ then find } \angle B$$

19. In any $\triangle ABC$, if $2 \cos B = \frac{a}{c}$

prove that the triangle is isosceles.

20. In a $\triangle ABC$, if $(a + b + c)(b + c - a) = \lambda bc$ then find the value of λ .

21. If the angles A, B, C of a triangle are in AP and its sides a, b, c are in GP, prove that a^2, b^2, c^2 are in AP.

22. If the line segment joining the points $P(a_1, b_1)$ and $Q(a_2, b_2)$ subtends an angle θ at the origin, prove that

$$\cos \theta = \left(\frac{a_1 a_2 + b_1 b_2}{\sqrt{a_1^2 + b_1^2} \sqrt{a_2^2 + b_2^2}} \right)$$

23. In a triangle ABC , if $\cot A, \cot B, \cot C$ are in AP, prove that a^2, b^2, c^2 are in AP.

24. If the sides of a triangle are a, b and $\sqrt{a^2 + ab + b^2}$ then find its greatest angle.

25. In a triangle ABC , if $a \cos A = b \cos B$, then prove that triangle is right angled isosceles.

26. In a triangle ABC , the angles are in AP, then prove that,

$$2 \cos \left(\frac{A - C}{2} \right) = \frac{a + c}{\sqrt{a^2 - ac + c^2}}$$

27. In a triangle ABC , prove that

$$\left(\frac{b^2 - c^2}{a^2} \right) \sin 2A + \left(\frac{c^2 - a^2}{b^2} \right) \sin 2B + \left(\frac{a^2 - b^2}{c^2} \right) \sin 2C = 0$$

28. In a triangle ABC , if $\angle A = 60^\circ$, then find the value of

$$\left(1 + \frac{a}{c} + \frac{b}{c} \right) \left(1 + \frac{c}{b} - \frac{a}{b} \right).$$

29. If $\frac{1}{a + c} + \frac{1}{b + c} = \frac{3}{a + b + c}$, then find $\angle C$

30. If in a triangle ABC ,

$$\frac{2 \cos A}{a} + \frac{2 \cos B}{b} + \frac{2 \cos C}{c} = \frac{1}{bc} + \frac{b}{ca},$$

then find the angle A in degrees.

PROJECTION RULE

31. In any $\triangle ABC$, prove that

$$2\left(a \sin^2 \frac{C}{2} + c \sin^2 \frac{A}{2}\right) = a + c - b$$

32. In any $\triangle ABC$, prove that

$$2\left(b \cos^2 \frac{C}{2} + c \cos^2 \frac{B}{2}\right) = a + b + c$$

33. In any $\triangle ABC$, prove that

$$(b+c) \cos A + (c+a) \cos B + (a+b) \cos C = (a+b+c)$$

34. In a $\triangle ABC$, prove that, $\frac{\sin B}{\sin C} = \frac{c-a \cos B}{b-a \cos C}$

35. In any $\triangle ABC$, prove that

$$2\left(a \sin^2 \frac{C}{2} + c \sin^2 \frac{A}{2}\right) = a + c - b$$

36. In any $\triangle ABC$, prove that

$$\frac{\cos A}{b \cos C + c \cos B} + \frac{\cos B}{c \cos A + a \cos C} + \frac{\cos C}{a \cos B + b \cos A} = \frac{a^2 + b^2 + c^2}{2abc}$$

37. In any $\triangle ABC$ prove that $2(bc \cos A + ca \cos B + ab \cos C) = (a^2 + b^2 + c^2)$

NAPIER ANALOGY

38. In any $\triangle ABC$, $b = \sqrt{3} + 1$, $c = \sqrt{3} - 1$

and $\angle A = 60^\circ$ then find the value of $\tan\left(\frac{B-C}{2}\right)$.

39. In any $\triangle ABC$, $b = \sqrt{3}$, $c = 1$, $B - C = 90^\circ$ then find $\angle A$.

40. If in a $\triangle ABC$, $a = 6$, $b = 3$ and $\cos(A-B) = \frac{4}{5}$, then find the angle C .

41. In a $\triangle ABC$, if

$$x = \tan\left(\frac{B-C}{2}\right) \tan\left(\frac{A}{2}\right),$$

$$y = \tan\left(\frac{C-A}{2}\right) \tan\left(\frac{B}{2}\right) \text{ and}$$

$$z = \tan\left(\frac{A-B}{2}\right) \tan\left(\frac{C}{2}\right),$$

then prove that $x + y + z + xyz = 0$

42. In a $\triangle ABC$, if $a = 5$, $b = 4$ and $\cos(A-B) = \frac{31}{32}$, then prove that $c = 6$.

HALF ANGLED FORMULA

43. In a $\triangle ABC$, if $a = 13$, $b = 14$ and $c = 15$, then find the value of

(i) $\sin \frac{A}{2}$

(ii) $\cos \frac{B}{2}$

(iii) $\cos A$

44. In a $\triangle ABC$, if $\cos\left(\frac{A}{2}\right) = \sqrt{\frac{b+c}{2c}}$, prove that $\triangle ABC$ is right angled at C .

45. In a $\triangle ABC$, prove that,

$$b \cos^2\left(\frac{C}{2}\right) + c \cos^2\left(\frac{B}{2}\right) = s$$

46. In a $\triangle ABC$, prove that

$$bc \cos^2\left(\frac{A}{2}\right) + ca \cos^2\left(\frac{B}{2}\right) + ab \cos^2\left(\frac{C}{2}\right) = s^2$$

47. In a $\triangle ABC$, prove that

$$2ac \sin\left(\frac{A-B+C}{2}\right) = (a^2 + c^2 - b^2).$$

48. In a $\triangle ABC$, $3a = b + c$,

then find the value of $\cot\left(\frac{B}{2}\right) \cot\left(\frac{C}{2}\right)$

49. In a $\triangle ABC$, prove that

$$1 - \tan\left(\frac{A}{2}\right) \tan\left(\frac{B}{2}\right) = \frac{2c}{(a+b+c)}$$

50. In a $\triangle ABC$, prove that:

$$\cot\left(\frac{A}{2}\right) + \cot\left(\frac{B}{2}\right) + \cot\left(\frac{C}{2}\right) = \left(\frac{a+b+c}{b+c-a}\right) \cot\left(\frac{A}{2}\right)$$

51. In a $\triangle ABC$, if $\cot\left(\frac{A}{2}\right)$, $\cot\left(\frac{B}{2}\right)$, $\cot\left(\frac{C}{2}\right)$ are in AP, then prove that a, b, c are in AP

52 In a $\triangle ABC$, $c(a+b) \cos \frac{B}{2} = b(a+c) \cos \frac{C}{2}$,

then prove that the triangle is isosceles.

53. In a $\triangle ABC$, prove that

$$\frac{\cot\left(\frac{A}{2}\right) + \cot\left(\frac{B}{2}\right) + \cot\left(\frac{C}{2}\right)}{\cot A + \cot B + \cot C} = \frac{(a+b+c)^2}{a^2 + b^2 + c^2}$$

54. In a $\triangle ABC$, if

$$c(a+b) \cos\left(\frac{B}{2}\right) = b(a+c) \cos\left(\frac{C}{2}\right)$$

then prove that the triangle ABC is isosceles.

AREA OF A TRIANGLE

55. In any $\triangle ABC$, if $a = \sqrt{2}$, $b = \sqrt{3}$

and $c = \sqrt{5}$, then find the area of the $\triangle ABC$.

56. In any $\triangle ABC$, prove that

$$\Delta = \frac{a^2 - b^2}{2} \times \frac{\sin A \sin B}{\sin(A - B)}$$

57. If the angles of triangle are 30° and 45° and the included side is $(\sqrt{3} + 1)$ cm., then prove that the area of the triangle is $\frac{1}{2}(\sqrt{3} + 1)$ cm².

58. In a $\triangle ABC$, prove that

$$\frac{\cot\left(\frac{A}{2}\right) + \cot\left(\frac{B}{2}\right) + \cot\left(\frac{C}{2}\right)}{\cot A + \cot B + \cot C} = \frac{(a + b + c)^2}{a^2 + b^2 + c^2}$$

59. If in a $\triangle ABC$, prove that $\Delta < \frac{s^2}{4}$.

60. If α, β, γ are the lengths of the altitudes of a $\triangle ABC$, then prove that

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{1}{\Delta}(\cot A + \cot B + \cot C)$$

61. If p_1, p_2, p_3 are the altitudes of a triangle from the vertices A, B, C and Δ be the area of the $\triangle ABC$, prove that

$$\frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} = \frac{2ab}{(a + b + c) \times \Delta} \times \cos^2\left(\frac{C}{2}\right).$$

62. If a, b, c and d are the sides of a quadrilateral, then find the minimum value of $\left(\frac{a^2 + b^2 + c^2}{d^2}\right)$

63. In a $\triangle ABC$, if $\cos A + \cos B + \cos C = \frac{3}{2}$, then the triangle is equilateral.

64. In a $\triangle PQR$, if $\sin P, \sin Q, \sin R$ are in AP then prove that its altitudes are in HP

65. In a $\triangle ABC$, $\Delta = (6 + 2\sqrt{3})$ sq.u

and $\angle B = 45^\circ, a = 2(\sqrt{3} + 1)$, then prove that the side b is 4

66. If the angles of a triangle are 30° and 45° and the included side is $(\sqrt{3} + 1)$, then prove that

$$\text{ar}(\triangle ABC) = \frac{1}{2}(\sqrt{3} + 1) \text{sq.u}$$

67. The two adjacent sides of a cyclic quadrilateral are 2 and 3 and the angle between them is 60° . If the area of the quadrilateral is $4\sqrt{3}$, then prove that the remaining two sides are 2 and 3 respectively.

M-N THEOREM

68. The median AD of a $\triangle ABC$ is perpendicular to AB . Prove that $\tan A + 2 \tan B = 0$

69. If D be the mid point of the side BC of the triangle ABC and Δ be its area, then prove that

$$\cot \theta = \frac{b^2 - c^2}{4\Delta}, \text{ where } \angle ADB = \theta$$

CIRCUM-CIRCLE AND CIRCUM-RADIUS

70. In a $\triangle ABC$, if $a = 18$ cm, $b = 24$ cm and $c = 30$ cm, then find its circum-radius

71. In an equilateral triangle of side $2\sqrt{3}$ cm, then find the circum-radius.

72. If the length of the sides of a triangle are 3, 4 and 5 units, then find its circum-radius R .

73. If $8R^2 = a^2 + b^2 + c^2$, then prove that the triangle is right angled

74. In any $\triangle ABC$, prove that $a \cos A + b \cos B + c \cos C = 4R \sin A \sin B \sin C$

75. In any $\triangle ABC$, prove that

$$D = 2R^2 \sin A \sin B \sin C$$

76. In any $\triangle ABC$, prove that,

$$\frac{\sin A}{a} + \frac{\sin B}{b} + \frac{\sin C}{c} = \frac{3}{2R}$$

77. In any $\triangle ABC$, a, b, c are in AP and p_1, p_2 and p_3 are the altitudes of the given triangle, then prove that,

$$\frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} \leq \frac{3R}{\Delta}.$$

78. If p_1, p_2 and p_3 are the altitudes of a $\triangle ABC$ from the vertices A, B and C respectively. then prove that

$$\frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} = \frac{1}{r}$$

79. If p_1, p_2 and p_3 are the altitudes of a $\triangle ABC$ from the vertices A, B and C respectively. then prove that

$$\frac{\cos A}{p_1} + \frac{\cos B}{p_2} + \frac{\cos C}{p_3} = \frac{1}{R}$$

80. In an acute angled $\triangle ABC$, prove that

$$\frac{\cos C}{\sqrt{4R^2 - c^2}} = \frac{1}{2R}$$

81. If p_1, p_2 and p_3 are the altitudes of a $\triangle ABC$ from the vertices A, B and C respectively. then prove that

$$p_1 p_2 p_3 = \frac{a^2 b^2 c^2}{8R^3}$$

82. If p_1, p_2 and p_3 are the altitudes of a $\triangle ABC$ from the vertices A, B and C respectively. then prove that

$$\frac{bp_1}{c} + \frac{cp_2}{a} + \frac{ap_3}{b} = \frac{a^2 + b^2 + c^2}{2R}$$

83. O is the circum-centre of $\triangle ABC$ and R_1, R_2 and R_3 are respectively the radii of the circum-centre of the triangles $\triangle OBC, \triangle OCA$ and $\triangle OAB$, prove that

$$\frac{a}{R_1} + \frac{b}{R_2} + \frac{c}{R_3} = \frac{abc}{R^3}$$

84. In an acute angled $\triangle ABC$, prove that

$$\frac{a \sec A + b \sec B + c \sec C}{\tan A \tan B \tan C} = 2R$$

85. In any $\triangle ABC$, prove that

$$(a \cos A + b \cos B + c \cos C) = 4R \sin A \sin B \sin C$$

IN-CIRCLE AND IN-RADIUS

86. In a $\triangle ABC$, if $a = 4$ cm, $b = 6$ cm and $c = 8$ cm, then find its in-radius.

87. If the sides of a triangle be 18, 24, 30 cm, then find its in-radius.

88. If the sides of a triangle are 3 : 7 : 8, then find $R : r$.

89. Two sides of a triangle are 2 and $\sqrt{3}$ and the included angle is 30° , then prove that its in-radius is $\frac{1}{2}(\sqrt{3} - 1)$.

90. In an equilateral triangle, prove that $R = 4r$

91. In a $\triangle ABC$, prove that

$$\frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} + \frac{1}{2rR}$$

92. In a $\triangle ABC$, prove that

$$\cos A + \cos B + \cos C = \left(1 + \frac{r}{R}\right).$$

93. In a $\triangle ABC$, prove that

$$\sin A + \sin B + \sin C = \frac{s}{R} = \frac{\Delta}{Rr}$$

94. In any $\triangle ABC$, prove that $a \cot A + b \cot B + c \cot C = 2(r + R)$

95. In a $\triangle ABC$, prove that

$$\frac{a \sec A + b \sec B + c \sec C}{2 \tan A \cdot \tan B \cdot \tan C} = R$$

96. In a $\triangle ABC$, prove that

$$(b + c) \tan \left(\frac{A}{2}\right) + (c + a) \tan \left(\frac{B}{2}\right) + (a + b) \tan \left(\frac{C}{2}\right) = 4(r + R)$$

97. In a $\triangle ABC$, if $C = 90^\circ$, prove that

$$\frac{1}{2}(a + b) = R + r$$

98. In any $\triangle ABC$, prove that

$$\cos^2 \left(\frac{A}{2}\right) + \cos^2 \left(\frac{B}{2}\right) + \cos^2 \left(\frac{C}{2}\right) = 2 + \frac{r}{2R}$$

99. If the distances of the sides of a triangle ABC from a circum-center be x, y and z respectively, then prove that

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = \frac{abc}{4xyz}$$

100. If in a $\triangle ABC$, O is the circum center and R is the circum-radius and R_1, R_2, R_3 are the circum radii of the triangles $\triangle OBC, \triangle OCA$ and $\triangle OAB$ respectively, then prove that

$$\frac{a}{R_1} + \frac{b}{R_2} + \frac{c}{R_3} = \frac{abc}{R^3}.$$

101. If p_1, p_2, p_3 are respectively the perpendiculars from the vertices of a $\triangle$ to the opposite sides, then prove that

$$p_1 \cdot p_2 \cdot p_3 = \frac{(abc)^2}{8R^3}.$$

102. Find The bisectors of the angles of a $\triangle ABC$

EXCIRCLE AND EX-RADII

103. In a $\triangle ABC$, if $a = 18$ cm, $b = 24$ cm, and $c = 30$ cm, then find the value of r_1, r_2 and r_3

104. In a triangle $\triangle ABC$, prove that $\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{r}$, where r is in radius and R_1, R_2, R_3 are exradii.

105. In a $\triangle ABC$, prove that

$$\frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3} = 0$$

106. In a $\triangle ABC$ if $\frac{s-c}{s-a} = \frac{b-c}{a-b}$, then prove that a, b, c are in AP

107. In a triangle if $\left(1 - \frac{r_1}{r_2}\right)\left(1 - \frac{r_1}{r_3}\right) = 2$, prove that the triangle is right angled.

108. In a triangle $\triangle ABC$, prove that $r_1 + r_2 + r_3 - r = 4R$

109. In a triangle $\triangle ABC$, prove that $r_1 r_2 + r_2 r_3 + r_3 r_1 = s^2$

110. In a triangle $\triangle ABC$, prove that $r_1 + r_2 - r_3 + r = 4R \cos C$

111. If r_1, r_2, r_3 are in HP, then prove that a, b, c are in AP.

112. In a triangle ABC , if a, b, c are in AP as well as in GP then prove that the value of $\left(\frac{r_1}{r_2} - \frac{r_2}{r_3} + 10\right)$ is 10.

113. In a triangle $\triangle ABC$, prove that

$$\frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} + \frac{1}{r^2} = \frac{a^2 + b^2 + c^2}{\Delta^2}$$

114. In a triangle $\triangle ABC$, prove that $(r_1 - r)(r_2 - r)(r_3 - r) = 4r^2 R$

115. If $r_1 < r_2 < r_3$ and the ex-radii of a right angled triangle and $r_1 = 1, r_2 = 2$, then prove that $r_3 = \frac{3 + \sqrt{17}}{2}$.

116. Two sides of a triangle are the roots of $x^2 - 5x + 3 = 0$. If the angle between the sides is $\frac{\pi}{3}$, then prove that the value of $r \cdot R$ is $2/3$.

117. In an isosceles triangle of which one angle is 120° , circle of radius $\sqrt{3}$ is inscribed, then prove that the area of the triangle is $(12 + 7\sqrt{3})$ sq. u.

118. If in a triangle $r = r_1 - r_2 - r_3$, then prove that the triangle is right angled.

119. In a $\triangle ABC$, prove that $r \cdot r_1 \cdot r_2 \cdot r_3 = \Delta^2$

120. Prove that $\frac{(r_1 + r_2)}{1 + \cos C} = \frac{(r_2 + r_3)}{1 + \cos A} = \frac{(r_3 + r_1)}{1 + \cos B}$.

121. Prove that $\left(\frac{1}{r} - \frac{1}{r_1}\right)\left(\frac{1}{r} - \frac{1}{r_2}\right)\left(\frac{1}{r} - \frac{1}{r_3}\right) = \frac{16R}{r^2(a+b+c)^2}$
122. Prove that $\left(\frac{1}{r_1} + \frac{1}{r_2}\right)\left(\frac{1}{r_2} + \frac{1}{r_3}\right)\left(\frac{1}{r_3} + \frac{1}{r_1}\right) = \frac{64R^3}{(abc)^2}$
123. In a $\triangle ABC$, prove that $r^2 + r_1^2 + r_2^2 + r_3^2 = 16R^2 - (a^2 + b^2 + c^2)$
124. In a $\triangle ABC$, prove that $\frac{(r_1 + r_2)(r_2 + r_3)(r_3 + r_1)}{(r_1 r_2 + r_2 r_3 + r_3 r_1)} = 4R$

REGULAR POLYGON

125. If $A_0, A_1, \dots, A_5$ be the consecutive vertices of a regular hexagon inscribed in a unit circle. Then find the product of length of $A_0 A_1, A_0 A_2$ and $A_0 A_4$.
126. If the Area of circle is A_1 and area of regular pentagon inscribed in the circle is A_2 . Find the ratio of area of two.
127. Let A_1, A_2, A_3, A_4 and A_5 be the vertices of a regular pentagon inscribed in a unit circle taken in order. Show that $A_1 A_2 \times A_1 A_3 = \sqrt{5}$.
128. The sides of a regular do-decagon is 2 ft. Find the radius of the circumscribed circle.
129. A regular pentagon and a regular decagon have the same perimeter. Find the ratio of its area.
130. If $2a$ be the sides of a regular polygon of n -sides. R and r be the circum-radius and inradius, then prove that $r + R = a \cot\left(\frac{\pi}{2n}\right)$.
131. A regular pentagon and a regular decagon have the same area, then find the ratio of their perimeter.
132. If the number of sides of two regular polygon having the same perimeter be n and $2n$ respectively, prove that their areas are in the ratio $2 \cos\left(\frac{\pi}{n}\right) : \left(1 + \cos\left(\frac{\pi}{n}\right)\right)$
133. Let $A_1, A_2, A_3, \dots, A_n$ be the vertices of an n -sided regular polygon such that $\frac{1}{A_1 A_2} = \frac{1}{A_1 A_3} + \frac{1}{A_1 A_4}$, then find the value of n .
134. If A, A_1, A_2, A_3 are the areas of incircle and the ex-circles of a triangle, then prove that $\frac{1}{\sqrt{A_1}} + \frac{1}{\sqrt{A_2}} + \frac{1}{\sqrt{A_3}} = \frac{1}{\sqrt{A}}$.
135. If the perimeter of the circle and the perimeter of the polygon of n -sides are same, then prove that the ratio of the area of the circle and the area of the polygon of n -sides is $\tan\left(\frac{\pi}{n}\right) : \frac{\pi}{n}$
136. Prove that the sum of the radii of the circle, which are respectively inscribed in and circum-scribed about a regular polygon of n -sides, is $\frac{a}{2} \cot\left(\frac{\pi}{2n}\right)$

(1)

$$\Rightarrow \frac{a^2 - b^2}{a^2 + b^2} = \frac{\sin^2 A - \sin^2 B}{\sin^2 C}$$

$$\Rightarrow \frac{a^2 - b^2}{a^2 + b^2} = \frac{k^2(a^2 - b^2)}{k^2 c^2}$$

$$\Rightarrow \frac{a^2 - b^2}{a^2 + b^2} = \frac{(a^2 - b^2)}{c^2}$$

$$\Rightarrow (a^2 - b^2) \left(\frac{1}{a^2 + b^2} - \frac{1}{c^2} \right) = 0$$

$$\Rightarrow (a^2 - b^2) = 0, \left(\frac{1}{a^2 + b^2} - \frac{1}{c^2} \right) = 0$$

$$\Rightarrow a^2 = b^2, \frac{1}{a^2 + b^2} = \frac{1}{c^2}$$

$$\Rightarrow a = b, a^2 + b^2 = c^2$$

Thus, the triangle is isosceles or right angled.

2. We have

$$\begin{aligned} \frac{\sin(B-C)}{\sin(B+C)} &= \frac{\sin(B-C)}{\sin(B+C)} \times \frac{\sin(B+C)}{\sin(B+C)} \\ &= \frac{\sin^2(B) - \sin^2(C)}{\sin^2(B+C)} \\ &= \frac{\sin^2(B) - \sin^2(C)}{\sin^2(\pi - A)} \\ &= \frac{\sin^2(B) - \sin^2(C)}{\sin^2(A)} \\ &= \frac{k^2 b^2 - k^2 c^2}{k^2 a^2} \\ &= \frac{b^2 - c^2}{a^2} \end{aligned}$$

3. We have

$$\begin{aligned} \angle C &= 180^\circ - (A + B) \\ &= 180^\circ - (45^\circ + 75^\circ) \\ &= 180^\circ - 120^\circ \\ &= 60^\circ \end{aligned}$$

From the sine rule, we can write

$$\begin{aligned} \frac{a}{\sin A} &= \frac{b}{\sin B} = \frac{c}{\sin C} \\ \Rightarrow \frac{a}{\sin(45^\circ)} &= \frac{b}{\sin(75^\circ)} = \frac{c}{\sin(60^\circ)} \\ \Rightarrow \frac{a}{\frac{1}{\sqrt{2}}} &= \frac{b}{\frac{\sqrt{3}+1}{2\sqrt{2}}} = \frac{c}{\frac{\sqrt{3}}{2}} = k(\text{say}) \end{aligned}$$

$$\text{Now, } a + c\sqrt{2}$$

$$= \frac{k}{\sqrt{2}} + \left(\frac{k\sqrt{3}}{2} \right) \sqrt{2}$$

$$= \frac{k}{\sqrt{2}} + \left(\frac{k\sqrt{3}}{\sqrt{2}} \right)$$

$$= \left(\frac{\sqrt{3}+1}{\sqrt{2}} \right) k$$

$$= 2 \left(\frac{\sqrt{3}+1}{2\sqrt{2}} \right) k$$

$$= 2b$$

4. Now,

$$\begin{aligned} \frac{a^2 \sin(B-C)}{\sin B + \sin C} &= \frac{ak \sin A \sin(B-C)}{\sin B + \sin C} \\ &= \frac{ak \sin(B+C) \sin(B-C)}{\sin B + \sin C} \\ &= ak \left(\frac{\sin^2(B) - \sin^2(C)}{\sin B + \sin C} \right) \\ &= ak(\sin B - \sin C) \\ &= k \sin A (\sin B - \sin C) \\ &= k(\sin A \sin B - \sin A \sin C) \end{aligned}$$

Similarly,

$$\frac{b^2 \sin(C-A)}{\sin C + \sin A} = k(\sin B \sin C - \sin A \sin B)$$

$$\text{and } \frac{c^2 \sin(A-B)}{\sin A + \sin B} = k(\sin A \sin C - \sin C \sin B)$$

Thus, LHS

$$\begin{aligned} &= k[\sin A \sin B - \sin A \sin C + \sin B \sin C \\ &\quad - \sin A \sin B + \sin A \sin C - \sin C \sin B] \\ &= 0 \end{aligned}$$

5. Now,

$$\begin{aligned} \frac{b^2 - c^2}{\cos B + \cos C} &= \frac{k^2(\sin^2 B - \sin^2 C)}{\cos B + \cos C} \\ &= \frac{k^2(1 - \cos^2 B - 1 + \cos^2 C)}{\cos B + \cos C} \\ &= -\frac{k^2(\cos^2 B - \cos^2 C)}{\cos B + \cos C} \\ &= -k^2(\cos B - \cos C) \end{aligned}$$

Similarly,

$$\frac{c^2 - a^2}{\cos C + \cos A} = -k^2(\cos C - \cos A)$$

$$\text{and } \frac{a^2 - b^2}{\cos A + \cos B} = -k^2(\cos A - \cos B)$$

Thus, LHS

$$= -k^2[\cos B - \cos C + \cos C - \cos A]$$

$$\begin{aligned} &\cos A - \cos B \\ &= 0 \end{aligned}$$

6. Given,

$$a^2, b^2, c^2 \text{ are in AP}$$

$$\Rightarrow b^2 - a^2 = c^2 - b^2$$

$$\Rightarrow \sin^2 B - \sin^2 A = \sin^2 C - \sin^2 B$$

$$\Rightarrow \sin(B+A) \sin(B-A) = \sin(C+B) \sin(C-B)$$

$$\Rightarrow \sin C(\sin B \cos A - \cos B \sin A) = \sin A(\sin C \cos B - \cos C \sin B)$$

Dividing both the sides by $\sin A \sin B \sin C$, we get,

$$\Rightarrow \cot A - \cot B = \cot B - \cot C$$

$$\Rightarrow \cot A, \cot B, \cot C \text{ are in AP}$$

7. We have

$$\cot\left(\frac{A}{2}\right) = \frac{b+c}{a}$$

$$\Rightarrow \cot\left(\frac{A}{2}\right) = \frac{\sin B + \sin C}{\sin A}$$

$$\Rightarrow \cot\left(\frac{A}{2}\right) = \frac{2 \sin\left(\frac{B+C}{2}\right) \cos\left(\frac{B-C}{2}\right)}{\sin A}$$

$$\Rightarrow \cot\left(\frac{A}{2}\right) = \frac{2 \sin\left(\frac{\pi}{2} - \frac{A}{2}\right) \cos\left(\frac{B-C}{2}\right)}{2 \sin\left(\frac{A}{2}\right) \cos\left(\frac{A}{2}\right)}$$

$$\Rightarrow \cot\left(\frac{A}{2}\right) = \frac{2 \cos\left(\frac{A}{2}\right) \cos\left(\frac{B-C}{2}\right)}{2 \sin\left(\frac{A}{2}\right) \cos\left(\frac{A}{2}\right)}$$

$$\Rightarrow \cot\left(\frac{A}{2}\right) = \frac{\cos\left(\frac{B-C}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\Rightarrow \frac{\cos\left(\frac{A}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\cos\left(\frac{B-C}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\Rightarrow \cos\left(\frac{A}{2}\right) = \cos\left(\frac{B-C}{2}\right)$$

$$\Rightarrow \frac{A}{2} = \frac{B-C}{2}$$

$$\Rightarrow A + C = B$$

$$\Rightarrow 2B = A + B + C = 180^\circ$$

$$\Rightarrow B = 90^\circ$$

Thus, the triangle is right angled.

8. Given,

$$a^2, b^2, c^2 \text{ are in AP.}$$

$$b^2 - a^2 = c^2 - b^2$$

$$\sin^2 B - \sin^2 C = \sin^2 C - \sin^2 B$$

$$\sin(B+A) \sin(B-A) = \sin(C+B) \sin(C-B)$$

$$\sin(C) \sin(B-A) = \sin(A) \sin(C-B)$$

$$\frac{\sin(B-A)}{\sin(C-B)} = \frac{\sin(A)}{\sin(C)}$$

$$\frac{\sin(A)}{\sin(C)} = \frac{\sin(A-B)}{\sin(B-C)}$$

9. We have

$$\prod \left(\frac{\sin^2 A + \sin A + 1}{\sin A} \right) = \left(\sin A + \frac{1}{\sin A} + 1 \right) \left(\sin B + \frac{1}{\sin B} + 1 \right) \left(\sin C + \frac{1}{\sin C} + 1 \right)$$

$$> (2+1)(2+1)(2+1) = 27$$

(applying AM $\geq$ GM)

10. Let $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$

We have,

$$\begin{aligned} \frac{b+c}{a} &= \frac{k(\sin B + \sin C)}{k \sin A} \\ &= \frac{(\sin B + \sin C)}{\sin A} \\ &= \frac{2 \sin\left(\frac{B+C}{2}\right) \cos\left(\frac{B-C}{2}\right)}{2 \sin\left(\frac{A}{2}\right) \cos\left(\frac{A}{2}\right)} \\ &= \frac{2 \sin\left(\frac{\pi}{2} - \frac{A}{2}\right) \cos\left(\frac{B-C}{2}\right)}{2 \sin\left(\frac{A}{2}\right) \cos\left(\frac{A}{2}\right)} \\ &= \frac{2 \cos\left(\frac{A}{2}\right) \cos\left(\frac{B-C}{2}\right)}{2 \sin\left(\frac{A}{2}\right) \cos\left(\frac{A}{2}\right)} \\ &= \frac{\cos\left(\frac{B-C}{2}\right)}{\sin\left(\frac{A}{2}\right)} \\ &= \frac{\sin\left(\frac{A}{2} + B\right)}{\sin\left(\frac{A}{2}\right)} \end{aligned}$$

11. Let $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$

We have, $\frac{a \sin(B-C)}{b^2 - c^2}$

$$\begin{aligned}
 &= \frac{k \sin A \cdot \sin (C-A)}{k^2 (\sin^2 C - \sin^2 A)} \\
 &= \frac{\sin (B+C) \cdot \sin (B-C)}{k (\sin^2 B - \sin^2 C)} \\
 &= \frac{(\sin^2 B - \sin^2 C)}{k (\sin^2 B - \sin^2 C)} = \frac{1}{k} \frac{b \sin (C-A)}{c^2 - a^2} \\
 &= \frac{k \sin (C+A) \cdot \sin (C-A)}{k^2 (\sin^2 C - \sin^2 A)} \\
 &= \frac{\sin^2 C - \sin^2 A}{k (\sin^2 C - \sin^2 A)} = \frac{1}{k}
 \end{aligned}$$

Also,

$$\begin{aligned}
 \frac{c \sin (A-B)}{a^2 - b^2} &= \frac{k \sin C \sin (A-B)}{k^2 (\sin^2 A - \sin^2 B)} \\
 &= \frac{\sin (A+B) \sin (A-B)}{k (\sin^2 A - \sin^2 B)} \\
 &= \frac{\sin^2 A - \sin^2 B}{k (\sin^2 A - \sin^2 B)} = \frac{1}{k}
 \end{aligned}$$

Hence, the result.

12. Given,

$$\begin{aligned}
 &\frac{1 + \cos (A-B) \cos C}{1 + \cos (A-C) \cos B} \\
 &= \frac{1 + \cos (A-B) \cos (\pi - (A+B))}{1 + \cos (A-C) \cos (\pi - (A+C))} \\
 &= \frac{1 - \cos (A-B) \cos (A+B)}{1 - \cos (A-C) \cos (A+C)} \\
 &= \frac{1 - \{\cos^2 A - \sin^2 B\}}{1 - \{\cos^2 A - \sin^2 C\}} \\
 &= \frac{\sin^2 A + \sin^2 B}{\sin^2 A + \sin^2 C} \\
 &= \frac{a^2 + b^2}{a^2 + c^2}
 \end{aligned}$$

13. Given,

$$\begin{aligned}
 &\cos A + 2 \cos B + \cos C = 2 \\
 \Rightarrow &\cos A + \cos C = 2(1 - \cos B) \\
 \Rightarrow &2 \cos \left(\frac{A+C}{2} \right) \cos \left(\frac{A-C}{2} \right) = 4 \sin^2 \left(\frac{B}{2} \right) \\
 \Rightarrow &\cos \left(\frac{\pi}{2} - \frac{B}{2} \right) \cos \left(\frac{A-C}{2} \right) = 2 \sin^2 \left(\frac{B}{2} \right) \\
 \Rightarrow &\sin \left(\frac{B}{2} \right) \cos \left(\frac{A-C}{2} \right) = 2 \sin^2 \left(\frac{B}{2} \right) \\
 \Rightarrow &\cos \left(\frac{A-C}{2} \right) = 2 \sin \left(\frac{B}{2} \right)
 \end{aligned}$$

Multiplying both sides by $2 \cos \left(\frac{B}{2} \right)$, we get,

$$\Rightarrow 2 \cos \left(\frac{B}{2} \right) \cos \left(\frac{A-C}{2} \right) = 2 \left(2 \cos \left(\frac{B}{2} \right) \sin \left(\frac{B}{2} \right) \right)$$

$$\begin{aligned}
 \Rightarrow &2 \sin \left(\frac{A+C}{2} \right) \cos \left(\frac{A-C}{2} \right) = 2 \sin B \\
 \Rightarrow &\sin A + \sin C = 2 \sin B \\
 \Rightarrow &a + c = 2b \\
 \Rightarrow &a, b, c \text{ are in AP}
 \end{aligned}$$

14 We have

$$\begin{aligned}
 &\cos A + \cos B + \sin A \sin B \sin C = 1 \\
 \Rightarrow &\frac{1 - \cos A \cos B}{\sin A \sin B} = \sin C \\
 \Rightarrow &\frac{1 - \cos A \cos B}{\sin A \sin B} = \sin C \leq 1 \\
 \Rightarrow &1 - \cos A \cos B \leq \sin A \sin B \\
 \Rightarrow &1 \leq \cos A \cos B + \sin A \sin B \\
 \Rightarrow &\cos (A-B) \geq 1 \\
 \Rightarrow &\cos (A-B) = 1 \\
 \Rightarrow &\cos (A-B) = \cos (0) \\
 \Rightarrow &A-B=0 \\
 \Rightarrow &A=B
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \sin C &= \frac{1 - \cos A \cos B}{\sin A \sin B} \\
 &= \frac{1 - \cos A \cos A}{\sin A \sin A} = \frac{1 - \cos^2 A}{\sin^2 A} = \frac{\sin^2 A}{\sin^2 A} = 1
 \end{aligned}$$

$$\Rightarrow C = 90^\circ$$

Hence, $A = 45^\circ = B$, $C = 90^\circ$

$$\begin{aligned}
 \text{Now, } \frac{a}{\sin A} &= \frac{b}{\sin B} = \frac{c}{\sin C} \\
 \Rightarrow \frac{a}{\sin 45^\circ} &= \frac{b}{\sin 45^\circ} = \frac{c}{\sin 90^\circ} \\
 \Rightarrow \frac{a}{1} &= \frac{b}{1} = \frac{c}{1} \\
 &\frac{a}{\sqrt{2}} = \frac{b}{\sqrt{2}} = \frac{c}{1} \\
 \Rightarrow \frac{a}{1} &= \frac{b}{1} = \frac{c}{\sqrt{2}}
 \end{aligned}$$

$$\Rightarrow a:b:c = 1:1:\sqrt{2}$$

Hence, the result.

15. We have,

$$\begin{aligned}
 &a(b \cos C - c \cos B) \\
 &= (ab \cos C - ac \cos B) \\
 &= ab \left(\frac{a^2 + b^2 - c^2}{2ab} \right) - ac \left(\frac{a^2 + c^2 - b^2}{2ac} \right) \\
 &= \left(\frac{a^2 + b^2 - c^2}{2} \right) - \left(\frac{a^2 + c^2 - b^2}{2} \right) \\
 &= \frac{1}{2} (a^2 + b^2 - c^2 - a^2 - c^2 + b^2) \\
 &= \frac{1}{2} (b^2 - c^2 - c^2 + b^2) \\
 &= \frac{1}{2} (2b^2 - 2c^2) \\
 &= (b^2 - c^2)
 \end{aligned}$$

Hence, the result.

16. We have $\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c}$

$$= \frac{1}{2abc}(b^2 + c^2 - a^2 + a^2 + c^2 - b^2 + a^2 + b^2 - c^2)$$

$$= \frac{(a^2 + b^2 + c^2)}{2abc}$$

17. We have

$$(a-b)^2 \cos^2\left(\frac{C}{2}\right) + (a+b)^2 \sin^2\left(\frac{C}{2}\right)$$

$$= (a^2 + b^2) \left(\cos^2\left(\frac{C}{2}\right) + \sin^2\left(\frac{C}{2}\right) \right)$$

$$- 2ab \left(\cos^2\left(\frac{C}{2}\right) - \sin^2\left(\frac{C}{2}\right) \right)$$

$$= (a+b)^2 - 2ab \cos C$$

$$= (a+b)^2 - 2ab \left(\frac{a^2 + b^2 - c^2}{2ab} \right)$$

$$= (a+b)^2 - (a^2 + b^2 - c^2)$$

$$= c^2$$

18. We have

$$(a+b+c)(a-b+c) = 3ac$$

$$\Rightarrow (a+c)^2 - b^2 = 3ac$$

$$\Rightarrow a^2 + c^2 - b^2 = 3ac - 2ac = ac$$

$$\Rightarrow \frac{a^2 + c^2 - b^2}{2ac} = \frac{ac}{2ac} = \frac{1}{2}$$

$$\Rightarrow \cos B = \frac{1}{2}$$

$$\Rightarrow B = \frac{\pi}{3}$$

Hence, the angle B is 60° .

19. We have $2 \cos B = \frac{a}{c}$

$$\Rightarrow 2 \left(\frac{a^2 + c^2 - b^2}{2ac} \right) = \frac{a}{c}$$

$$\Rightarrow \left(\frac{a^2 + c^2 - b^2}{a} \right) = a$$

$$\Rightarrow (a^2 + c^2 - b^2) = a^2$$

$$\Rightarrow (c^2 - b^2) = 0$$

$$\Rightarrow c^2 = b^2$$

$$\Rightarrow c = b$$

Thus, the triangle is isosceles.

20. Given, $(a+b+c)(b+c-a) = \lambda bc$

$$\{(b+c)^2 - a^2\} = \lambda bc$$

$$(b^2 + c^2 - a^2) = (\lambda - 2)bc$$

$$\frac{(b^2 + c^2 - a^2)}{2bc} = \frac{(\lambda - 2)bc}{2bc} = \frac{\lambda - 2}{2}$$

$$\cos A = \frac{\lambda - 2}{2}$$

$$-1 \leq \frac{\lambda - 2}{2} \leq 1$$

$$-2 \leq \lambda - 2 \leq 2$$

$$0 \leq \lambda \leq 4$$

21. Given, A, B, C are in AP

$$2B = A + C$$

$$3B = A + B + C = \pi$$

$$B = \frac{\pi}{3}$$

$$\cos B = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

$$\frac{a^2 + c^2 - b^2}{2ac} = \frac{1}{2}$$

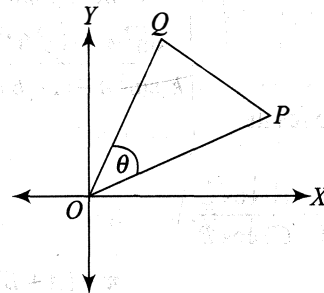
$$a^2 + c^2 - b^2 = ac$$

$$a^2 + c^2 - b^2 = b^2$$

$$2b^2 = a^2 + c^2$$

$$a^2, b^2, c^2 \in \text{AP}$$

22. Given, O be the origin and $\angle POQ = \theta$



Now, $\cos \theta = \frac{OP^2 + OQ^2 - PQ^2}{2OP \cdot OQ}$

$$\cos \theta = \frac{(a_1^2 + b_1^2) + (a_2^2 + b_2^2) - \{(a_1 - a_2)^2 + (b_1 - b_2)^2\}}{2\sqrt{a_1^2 + b_1^2} \cdot \sqrt{a_2^2 + b_2^2}}$$

$$= \frac{2(a_1 a_2 + b_1 b_2)}{2\sqrt{a_1^2 + b_1^2} \cdot \sqrt{a_2^2 + b_2^2}}$$

$$= \frac{(a_1 a_2 + b_1 b_2)}{\sqrt{a_1^2 + b_1^2} \cdot \sqrt{a_2^2 + b_2^2}}$$

23. Given, $\cot A, \cot B, \cot C$ are in AP

$$\frac{\cos A}{\sin A}, \frac{\cos B}{\sin B}, \frac{\cos C}{\sin C} \in \text{AP}$$

$$\frac{b^2 + c^2 - a^2}{2abck}, \frac{a^2 + c^2 - b^2}{2abck}, \frac{a^2 + b^2 - c^2}{2abck} \in \text{AP}$$

$$(b^2 + c^2 - a^2), (a^2 + c^2 - b^2), (a^2 + b^2 - c^2) \in \text{AP}$$

Subtracting $(a^2 + b^2 + c^2)$ to each term

$$(-2a^2), (-2b^2), (-2c^2) \in \text{AP}$$

$$a^2, b^2, c^2 \in \text{AP}$$

24. Let $c = \sqrt{a^2 + ab + b^2}$

Clearly, side c is the greatest

Thus, the angle C is the greatest.

$$\begin{aligned}\text{Now, } \cos C &= \frac{a^2 + b^2 - c^2}{2ab} \\ &= \frac{a^2 + b^2 - (a^2 + ab + b^2)}{2ab} \\ &= -\frac{ab}{2ab} = -\frac{1}{2}\end{aligned}$$

$$\text{Thus, } \angle C = \frac{2\pi}{3}$$

25. We have $a \cos A = b \cos B$

$$\begin{aligned}a \left(\frac{b^2 + c^2 - a^2}{2bc} \right) &= b \left(\frac{a^2 + c^2 - b^2}{2ac} \right) \\ \Rightarrow a \left(\frac{b^2 + c^2 - a^2}{b} \right) &= b \left(\frac{a^2 + c^2 - b^2}{a} \right) \\ \Rightarrow a^2(b^2 + c^2 - a^2) &= b^2(a^2 + c^2 - b^2) \\ \Rightarrow a^2(c^2 - a^2) &= b^2(c^2 - b^2) \\ \Rightarrow c^2(a^2 - b^2) &= (a^4 - b^4) \\ \Rightarrow c^2(a^2 - b^2) &= (a^2 - b^2)(a^2 + b^2) \\ \Rightarrow (a^2 - b^2)((a^2 + b^2) - c^2) &= 0 \\ \Rightarrow (a^2 - b^2) = 0, (a^2 + b^2) &= c^2 \\ \Rightarrow a = b, (a^2 + b^2) &= c^2\end{aligned}$$

Thus, the triangle is right angled isosceles.

26. Since the angles are in AP, so $A + C = 2B$

$$\begin{aligned}\Rightarrow A + B + C &= 3B \\ \Rightarrow 3B &= 180^\circ \\ \Rightarrow B &= 60^\circ \\ \Rightarrow \cos B &= \cos(60^\circ) = \frac{1}{2}\end{aligned}$$

$$\Rightarrow \frac{a^2 + c^2 - b^2}{2ac} = \frac{1}{2}$$

$$\begin{aligned}\Rightarrow a^2 + c^2 - b^2 &= ac \\ \Rightarrow a^2 + c^2 - ac &= b^2\end{aligned}$$

Now, RHS

$$\begin{aligned}&= \frac{a+c}{\sqrt{a^2 - ac + c^2}} \\ &= \frac{a+c}{b} \\ &= \frac{k(\sin A + \sin C)}{k \sin B} \\ &= \frac{2 \sin \left(\frac{A+C}{2} \right) \cos \left(\frac{A-C}{2} \right)}{2 \sin \left(\frac{B}{2} \right) \cos \left(\frac{B}{2} \right)} \\ &= \frac{\cos \left(\frac{B}{2} \right) \cos \left(\frac{A-C}{2} \right)}{\sin \left(\frac{B}{2} \right) \cos \left(\frac{B}{2} \right)}\end{aligned}$$

$$\begin{aligned}&= \frac{\cos \left(\frac{A-C}{2} \right)}{\sin \left(\frac{B}{2} \right)} \\ &= \frac{\cos \left(\frac{A-C}{2} \right)}{\sin \left(\frac{60^\circ}{2} \right)} \\ &= 2 \cos \left(\frac{A-C}{2} \right)\end{aligned}$$

27. We have

$$\begin{aligned}\left(\frac{b^2 - c^2}{a^2} \right) \sin 2A &= \left(\frac{b^2 - c^2}{a^2} \right) (2 \sin A \cdot \cos A) \\ &= \left(\frac{b^2 - c^2}{a^2} \right) \left(2ka \left(\frac{b^2 + c^2 - a^2}{2bc} \right) \right) \\ &= \left(\frac{(b^2 - c^2)(b^2 + c^2 - a^2)}{kabc} \right) \\ &= \frac{1}{kabc} \times \{(b^4 - c^4) - a^2(b^2 - c^2)\}\end{aligned}$$

Similarly,

$$\begin{aligned}\left(\frac{c^2 - a^2}{b^2} \right) \times \sin 2B &= \frac{1}{kabc} \times \{(c^4 - a^4) - b^2(c^2 - a^2)\} \\ &\quad \left(\frac{a^2 - b^2}{c^2} \right) \sin 2C \text{ and} \\ &= \frac{1}{kabc} \times \{(a^4 - b^4) - c^2(a^2 - b^2)\}\end{aligned}$$

$$\begin{aligned}\text{Thus, } \left(\frac{b^2 - c^2}{a^2} \right) \sin 2A &+ \left(\frac{c^2 - a^2}{b^2} \right) \sin 2B \\ &+ \left(\frac{a^2 - b^2}{c^2} \right) \sin 2C\end{aligned}$$

$$\begin{aligned}&= \frac{1}{kabc} \times \{(b^4 - c^4) - a^2(b^2 - c^2)\} \\ &+ \frac{1}{kabc} \times \{(c^4 - a^4) - b^2(c^2 - a^2)\} \\ &+ \frac{1}{kabc} \times \{(a^4 - b^4) - c^2(a^2 - b^2)\} \\ &= \frac{1}{kabc} \times [(b^4 - c^4 + c^4 - a^4 + a^4 - b^4) \\ &= a^2(c^2 - b^2) + b^2(a^2 - c^2) + c^2(b^2 - a^2)] \\ &= 0\end{aligned}$$

28. Given,

$$\begin{aligned}\angle A &= 60^\circ \\ \Rightarrow \cos A &= \cos(60^\circ) \\ \Rightarrow \cos A &= \frac{1}{2}\end{aligned}$$

$$\Rightarrow \frac{b^2 + c^2 - a^2}{2bc} = \frac{1}{2}$$

$$\Rightarrow (b^2 + c^2 - a^2) = b \quad \dots(i)$$

Now,

$$\left(1 + \frac{a}{c} + \frac{b}{c}\right) \left(1 + \frac{c}{b} - \frac{a}{b}\right) = \left(\frac{c+a+b}{c}\right) \left(\frac{b+c-a}{b}\right)$$

$$= \left(\frac{b+c+a}{c}\right) \left(\frac{b+c-a}{b}\right)$$

$$= \left(\frac{(b+c)^2 - a^2}{bc}\right)$$

$$= \left(\frac{b^2 + c^2 - a^2 + 2bc}{bc}\right)$$

$$= \left(\frac{bc + 2bc}{bc}\right), \text{ from (i)}$$

$$= \left(\frac{3bc}{bc}\right)$$

$$= 3$$

29. Given, $\frac{1}{a+c} + \frac{1}{b+c} = \frac{3}{a+b+c}$

$$\Rightarrow \frac{a+b+c}{a+c} + \frac{a+b+c}{b+c} = 3$$

$$\Rightarrow 1 + \frac{b}{a+c} + 1 + \frac{a}{b+c} = 3$$

$$\Rightarrow \frac{b}{a+c} + \frac{a}{b+c} = 1$$

$$\Rightarrow b(b+c) + a(a+c) = (a+c)(b+c)$$

$$\Rightarrow b^2 + bc + a^2 + ac = ab + ac + bc + c^2$$

$$\Rightarrow a^2 + b^2 - c^2 = ab$$

$$\text{Now, } \cos(C) = \left(\frac{a^2 + b^2 - c^2}{2ab}\right)$$

$$= \frac{ab}{2ab} = \frac{1}{2}$$

$$\Rightarrow C = \frac{\pi}{3}$$

30. We have

$$\frac{2 \cos A}{a} + \frac{2 \cos B}{b} + \frac{2 \cos C}{c} = \frac{1}{bc} + \frac{b}{ca}$$

$$\Rightarrow \frac{2bc \cos A}{abc} + \frac{ac \cos B}{abc} + \frac{2bc \cos C}{abc}$$

$$= \frac{a^2}{abc} + \frac{b^2}{abc}$$

$$\Rightarrow 2bc \cos A + ac \cos B + 2bc \cos C$$

$$= a^2 + b^2$$

$$\Rightarrow (b^2 + c^2 - a^2) + \frac{1}{2}(a^2 + c^2 - b^2)$$

$$+ (a^2 + b^2 - c^2) = a^2 + b^2$$

$$\Rightarrow 2b^2 - 2a^2 + c^2 + a^2 - b^2 = 0$$

$$\Rightarrow a^2 = b^2 + c^2$$

ΔABC is a right angled triangle at A

Thus, $\angle A = 90^\circ$

31. We have

$$2 \left(a \sin^2 \left(\frac{C}{2} \right) + c \sin^2 \left(\frac{A}{2} \right) \right)$$

$$= \left(2a \sin^2 \left(\frac{C}{2} \right) + 2c \sin^2 \left(\frac{A}{2} \right) \right)$$

$$= a(1 - \cos C) + c(1 - \cos A)$$

$$= a + c - (a \cos C + c \cos A)$$

$$= (a + c - b)$$

32. We have

$$2 \left(b \cos^2 \left(\frac{C}{2} \right) + c \cos^2 \left(\frac{B}{2} \right) \right)$$

$$= \left(2b \cos^2 \left(\frac{C}{2} \right) + 2c \cos^2 \left(\frac{B}{2} \right) \right)$$

$$= b(1 + \cos C) + c(1 + \cos B)$$

$$= b + c + (b \cos C + c \cos B)$$

$$= (b + c + a)$$

$$= (a + b + c)$$

33. We have

$$(b+c) \cos A + (c+a) \cos B + (a+b) \cos C$$

$$= (b \cos A + a \cos B) + (c \cos A + a \cos C)$$

$$+ (b \cos C + c \cos B)$$

$$= (c + b + a)$$

$$= (a + b + c)$$

34. We have

$$\frac{c - a \cos B}{b - a \cos C} = \frac{a \cos B + b \cos A - a \cos B}{c \cos A + a \cos C - a \cos C}$$

$$= \frac{b \cos A}{c \cos A}$$

$$= \frac{b}{c}$$

$$= \frac{k \sin B}{k \sin C}$$

$$= \frac{\sin B}{\sin C}$$

35. We have

$$2 \left(a \sin^2 \frac{C}{2} + c \sin^2 \frac{A}{2} \right)$$

$$= a \left(2 \sin^2 \left(\frac{C}{2} \right) \right) + c \left(2 \sin^2 \left(\frac{A}{2} \right) \right)$$

$$= a(1 - \cos C) + c(1 - \cos A)$$

$$= a + c - (a \cos C + c \cos A)$$

$$= a + c - b$$

36. We have

$$\begin{aligned} & \frac{\cos A}{b \cos C + c \cos B} + \frac{\cos B}{c \cos A + a \cos C} \\ & + \frac{\cos C}{a \cos B + b \cos A} \\ & = \frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} \\ & = \left(\frac{b^2 + c^2 - a^2}{2abc} \right) + \left(\frac{c^2 + a^2 - b^2}{2abc} \right) \\ & \quad + \left(\frac{a^2 + b^2 - c^2}{2abc} \right) \\ & = \frac{1}{2abc} \{ (b^2 + c^2 - a^2) \\ & \quad + (c^2 + a^2 - b^2) + (a^2 + b^2 - c^2) \} \\ & = \frac{(a^2 + b^2 + c^2)}{2abc} \end{aligned}$$

37. We have

$$\begin{aligned} & 2(bc \cos A + ca \cos B + ab \cos C) \\ & = 2bc \cos A + 2ca \cos B + 2ab \cos C \\ & = 2bc \left(\frac{b^2 + c^2 - a^2}{2bc} \right) + 2ca \left(\frac{a^2 + c^2 - b^2}{2ac} \right) \\ & \quad + 2ab \left(\frac{a^2 + b^2 - c^2}{2ab} \right) \\ & = (b^2 + c^2 - a^2) + (a^2 + c^2 - b^2) + (a^2 + b^2 - c^2) \\ & = (a^2 + b^2 + c^2) \end{aligned}$$

38. As we know that,

$$\begin{aligned} \tan \left(\frac{B-C}{2} \right) &= \left(\frac{b-c}{b+c} \right) \cot \left(\frac{A}{2} \right) \\ &= \left(\frac{(\sqrt{3}+1) - (\sqrt{3}-1)}{(\sqrt{3}+1) + (\sqrt{3}-1)} \right) \cot \left(\frac{60^\circ}{2} \right) \\ &= \frac{1}{\sqrt{3}} \cot(30^\circ) \\ &= \frac{1}{\sqrt{3}} \times \sqrt{3} \\ &= 1 \end{aligned}$$

39. As we know that,

$$\begin{aligned} \tan \left(\frac{B-C}{2} \right) &= \left(\frac{b-c}{b+c} \right) \cot \left(\frac{A}{2} \right) \\ \Rightarrow \tan \left(\frac{90^\circ}{2} \right) &= \left(\frac{\sqrt{3}-1}{\sqrt{3}+1} \right) \cot \left(\frac{A}{2} \right) \\ \Rightarrow \tan(45^\circ) &= (2-\sqrt{3}) \cot \left(\frac{A}{2} \right) \\ \Rightarrow \cot \left(\frac{A}{2} \right) &= \frac{1}{(2-\sqrt{3})} = (2+\sqrt{3}) \\ \Rightarrow \cot \left(\frac{A}{2} \right) &= \cot(15^\circ) \end{aligned}$$

$$\Rightarrow A = 30^\circ$$

Hence, the angle A is 30° .

40. Given,

$$\begin{aligned} \cos(A-B) &= \frac{4}{5} \\ \Rightarrow 2 \cos^2 \left(\frac{A-B}{2} \right) - 1 &= \frac{4}{5} \\ \Rightarrow 2 \cos^2 \left(\frac{A-B}{2} \right) &= 1 + \frac{4}{5} = \frac{9}{5} \\ \Rightarrow \cos^2 \left(\frac{A-B}{2} \right) &= \frac{9}{10} \\ \Rightarrow \cos \left(\frac{A-B}{2} \right) &= \frac{3}{\sqrt{10}} \\ \Rightarrow \tan \left(\frac{A-B}{2} \right) &= \frac{1}{3} \\ \Rightarrow \left(\frac{a-b}{a+b} \right) \cot \left(\frac{C}{2} \right) &= \frac{1}{3} \\ \Rightarrow \left(\frac{6-3}{6+3} \right) \cot \left(\frac{C}{2} \right) &= \frac{1}{3} \\ \Rightarrow \frac{1}{3} \cot \left(\frac{C}{2} \right) &= \frac{1}{3} \\ \Rightarrow \cot \left(\frac{C}{2} \right) &= 1 \\ \Rightarrow \frac{C}{2} &= \frac{\pi}{4} \\ \Rightarrow C &= \frac{\pi}{2} \end{aligned}$$

Hence, the value of C is $\frac{\pi}{2}$.

43. We have

$$\begin{aligned} 2s &= a + b + c = 13 + 14 + 15 \\ s &= \frac{13 + 14 + 15}{2} = \frac{42}{2} = 21 \end{aligned}$$

$$\begin{aligned} \text{(i)} \quad \sin \frac{A}{2} &= \sqrt{\frac{(s-b)(s-c)}{bc}} \\ &= \sqrt{\frac{(21-14)(21-15)}{14 \cdot 15}} \\ &= \sqrt{\frac{7 \cdot 6}{14 \cdot 15}} = \frac{1}{\sqrt{5}} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \cos \frac{A}{2} &= \sqrt{\frac{s(s-a)}{bc}} \\ &= \sqrt{\frac{21(21-13)}{14 \cdot 15}} \\ &= \sqrt{\frac{21 \cdot 8}{14 \cdot 15}} \\ &= \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}} \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad \cos A &= 2 \cos^2 \left(\frac{A}{2} \right) - 1 \\
 &= 2 \left(\frac{4}{5} \right) - 1 \\
 &= \frac{8-5}{5} = \frac{3}{5}
 \end{aligned}$$

$$44. \text{ We have } \cos \left(\frac{A}{2} \right) = \sqrt{\frac{b+c}{2c}}$$

$$\Rightarrow \sqrt{\frac{s(s-a)}{bc}} = \sqrt{\frac{b+c}{2c}}$$

$$\Rightarrow \frac{s(s-a)}{bc} = \frac{b+c}{2c}$$

$$\Rightarrow 2s(2s-2a) = 2b(b+c)$$

$$\Rightarrow (a+b+c)(b+c-a) = 2b(b+c)$$

$$\Rightarrow ((b+c)^2 - a^2) = 2b(b+c)$$

$$\Rightarrow b^2 + c^2 - a^2 = 2b^2$$

$$\Rightarrow a^2 + b^2 = c^2$$

Thus, the triangle ABC is right angled at C .

45. We have

$$\begin{aligned}
 &b \cos^2 \left(\frac{C}{2} \right) + c \cos^2 \left(\frac{B}{2} \right) \\
 &= b \left(\frac{s(s-c)}{ab} \right) + c \left(\frac{s(s-b)}{ac} \right) \\
 &= \frac{s}{a} (s-c + s-b) \\
 &= \frac{s}{a} (2s - c - b) \\
 &= \frac{s}{a} (a + b + c - c - b) \\
 &= s
 \end{aligned}$$

46. We have

$$\begin{aligned}
 &bc \cos^2 \left(\frac{A}{2} \right) + ca \cos^2 \left(\frac{B}{2} \right) + ab \cos^2 \left(\frac{C}{2} \right) \\
 &= bc \left(\frac{s(s-a)}{bc} \right) + ca \left(\frac{s(s-b)}{ca} \right) + ab \left(\frac{s(s-c)}{ab} \right) \\
 &= s(s-a) + s(s-b) + s(s-c) \\
 &= s(3s - (a+b+c)) \\
 &= s(3s - 2s) \\
 &= s \times s \\
 &= s^2
 \end{aligned}$$

$$47. \text{ We have } 2ac \sin \left(\frac{A-B+C}{2} \right)$$

$$= 2ac \sin \left(\frac{A+C-B}{2} \right)$$

$$= 2ac \sin \left(\frac{\pi - B - B}{2} \right)$$

$$= 2ac \sin \left(\frac{\pi}{2} - B \right)$$

$$= 2ac \cos B$$

$$= 2ac \left(\frac{a^2 + c^2 - b^2}{2ac} \right)$$

$$= (a^2 + c^2 - b^2)$$

$$48. \text{ We have } \cot \left(\frac{B}{2} \right) \cot \left(\frac{C}{2} \right)$$

$$= \sqrt{\frac{s(s-b)}{(s-a)(s-c)}} \times \sqrt{\frac{s(s-c)}{(s-a)(s-b)}}$$

$$= \sqrt{\frac{s(s-b)}{(s-a)(s-c)}} \times \frac{s(s-c)}{(s-a)(s-b)}$$

$$= \sqrt{\frac{s^2}{(s-a)^2}}$$

$$= \frac{s}{s-a}$$

$$= \frac{2s}{2s-2a}$$

$$= \frac{a+b+c}{a+b+c-2a}$$

$$= \frac{4a}{4a-2a}$$

$$= 2$$

$$49. \text{ We have } 1 - \tan \left(\frac{A}{2} \right) \tan \left(\frac{B}{2} \right)$$

$$= 1 - \frac{\sqrt{\frac{s-b}{s(s-a)}} \sqrt{\frac{s-c}{s(s-b)}}}{\sqrt{\frac{s-b}{s(s-a)}} \sqrt{\frac{s-c}{s(s-b)}}}$$

$$= 1 - \frac{(s-c)^2}{s^2}$$

$$= 1 - \frac{(s-c)^2}{s^2}$$

$$= \frac{c}{s}$$

$$= \frac{2c}{2s}$$

$$= \frac{2c}{(a+b+c)}$$

Hence, the result.

$$50. \text{ We have } \cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2}$$

$$= \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} + \sqrt{\frac{s(s-b)}{(s-a)(s-c)}} + \sqrt{\frac{s(s-c)}{(s-a)(s-b)}}$$

$$= \sqrt{\frac{s^2(s-a)^2}{s(s-a)(s-b)(s-c)}} + \sqrt{\frac{s^2(s-b)^2}{s(s-b)(s-a)(s-c)}}$$

$$+ \sqrt{\frac{s^2(s-c)^2}{s(s-c)(s-a)(s-b)}}$$

$$\begin{aligned}
 &= \sqrt{\frac{s^2(s-a)^2}{\Delta^2}} + \sqrt{\frac{s^2(s-b)^2}{\Delta^2}} + \sqrt{\frac{s^2(s-c)^2}{\Delta^2}} \\
 &= \frac{s(s-a)}{\Delta} + \frac{s(s-b)}{\Delta} + \frac{s(s-c)}{\Delta} \\
 &= \frac{s}{\Delta}(s-a+s-b+s-c) \\
 &= \frac{s}{\Delta}(3s-(a+b+c)) \\
 &= \frac{s}{\Delta}(3s-2s) \\
 &= \frac{s^2}{\Delta} \\
 &= \frac{s^2}{\sqrt{s(s-a)(s-b)(s-c)}} \\
 &= \frac{s}{(s-a)} \times \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} \\
 &= \frac{2s}{(2s-2a)} \times \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} \\
 &= \frac{(a+b+c)}{(b+c-a)} \times \cot\left(\frac{A}{2}\right)
 \end{aligned}$$

51. Given, $\cot \frac{A}{2}, \cot \frac{B}{2}, \cot \frac{C}{2}$ are in AP

$$\begin{aligned}
 \Rightarrow 2 \cot\left(\frac{B}{2}\right) &= \cot\left(\frac{A}{2}\right) + \cot\left(\frac{C}{2}\right) \\
 \Rightarrow 2 \sqrt{\frac{s(s-b)}{(s-a)(s-c)}} &= \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} + \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} \\
 \Rightarrow 2 \sqrt{\frac{(s-b)}{(s-a)(s-c)}} &= \sqrt{\frac{(s-a)}{(s-b)(s-c)}} + \sqrt{\frac{(s-c)}{(s-a)(s-b)}} \\
 \Rightarrow 2 \sqrt{\frac{(s-b)^2}{(s-b)(s-a)(s-c)}} &= \sqrt{\frac{(s-a)^2}{(s-a)(s-b)(s-c)}} + \sqrt{\frac{(s-c)^2}{(s-a)(s-b)(s-c)}} \\
 \Rightarrow 2(s-b) &= (s-a) + (s-c) \\
 \Rightarrow 2(s-b) &= (2s-a-c) \\
 \Rightarrow 2b &= a+c \\
 \Rightarrow a, b, c &\in \text{AP}
 \end{aligned}$$

52. We have $c(a+b) \cos \frac{B}{2} = b(a+c) \cos \frac{C}{2}$

$$\begin{aligned}
 \Rightarrow c(a+b) \times \sqrt{\frac{s(s-b)}{ac}} &= b(a+c) \times \sqrt{\frac{s(s-c)}{ab}} \\
 \Rightarrow c(a+b) \times \sqrt{\frac{(s-b)}{c}} &= b(a+c) \times \sqrt{\frac{(s-c)}{b}} \\
 \Rightarrow c^2(a+b)^2 \times \frac{(s-b)}{c} &= b^2(a+c)^2 \times \frac{(s-c)}{b} \\
 \Rightarrow c(a+b)^2 \times (s-b) &= b(a+c)^2 \times (s-c) \\
 \Rightarrow c(a+b)^2 \times (s-b) &= b(a+c)^2 \times (s-c) \\
 \Rightarrow c(a+b)^2 \times (a+c+b) &= b(a+c)^2 \times (a+b-c) \\
 \Rightarrow \frac{(a+c-b)}{b(a+c)^2} &= \frac{(a+b-c)}{c(a+b)^2} \\
 \Rightarrow \frac{1}{b(a+c)} \frac{1}{(a+c)^2} &= \frac{1}{c(a+b)} \frac{1}{(a+b)^2} \\
 \Rightarrow \frac{1}{b(a+c)} \frac{1}{c(a+b)} &= \frac{1}{(a+c)^2} \frac{1}{(a+b)^2} \\
 \Rightarrow \frac{ac+c^2-ab-b^2}{bc(a+b)(a+c)} &= \frac{(a+b)^2-(a+c)^2}{(a+c)^2(a+b)^2} \\
 \Rightarrow \frac{a(c-b)+(c^2-b^2)}{bc} &= \frac{2a(b-c)+(b^2-c^2)}{(a+c)(a+b)} \\
 \Rightarrow \frac{(c-b)(a+b+c)}{bc} &= \frac{(b-c)(2a+b+c)}{(a+c)(a+b)} \\
 \Rightarrow (c-b) \left(\frac{(a+b+c)}{bc} + \frac{(2a+b+c)}{(a+c)(a+b)} \right) &= 0 \\
 \Rightarrow (c-b) &= 0 \\
 \Rightarrow \left(\frac{(a+b+c)}{bc} + \frac{(2a+b+c)}{(a+c)(a+b)} \right) &\neq 0 \\
 \Rightarrow b &= c \\
 \Rightarrow \Delta &\text{ is isosceles}
 \end{aligned}$$

55. $\Delta ABC = \frac{1}{2} \times \sqrt{2} \times \sqrt{3} = \frac{\sqrt{3}}{2}$ s.u.

56. We have

$$\begin{aligned}
 \frac{a^2-b^2}{2} \times \frac{\sin A \sin B}{\sin(A-B)} &= \frac{k^2(\sin^2 A - \sin^2 B)}{2} \times \frac{\sin A \cdot \sin B}{\sin(A-B)} \\
 &= \frac{k^2 \times \sin(A+B) \times \sin(A-B)}{2} \times \frac{\sin A \cdot \sin B}{\sin(A-B)} \\
 &= \frac{k^2 \times \sin(A+B) \times \sin A \cdot \sin B}{2} \\
 &= \frac{k^2 \times \sin(\pi - C) \times \sin A \cdot \sin B}{2}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{k^2 \times \sin C \times \frac{a}{k} \times \frac{b}{k}}{2} \\
 &= \frac{1}{2} \times ab \sin C \\
 &= \Delta
 \end{aligned}$$

57. As we know that, $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

$$\Rightarrow \frac{(\sqrt{3}+1)}{\sin(105^\circ)} = \frac{b}{\sin 45^\circ} = \frac{c}{\sin 30^\circ}$$

$$\Rightarrow \frac{(\sqrt{3}+1)}{\cos(15^\circ)} = \frac{b}{\sin 45^\circ} = \frac{c}{\sin 30^\circ}$$

$$\Rightarrow \frac{(\sqrt{3}+1)}{\frac{\sqrt{3}+1}{2\sqrt{2}}} = \frac{b}{\frac{1}{\sqrt{2}}} = \frac{c}{\frac{1}{2}}$$

$$\Rightarrow 2\sqrt{2} = b\sqrt{2} = 2c$$

$$\Rightarrow b = 2 \text{ and } c = \sqrt{2}$$

Hence, the area of the triangle is $= \frac{1}{2}bc \sin A$

$$= \frac{1}{2} \times 2 \times \sqrt{2} \times \sin(105^\circ)$$

$$= \frac{1}{2} \times 2 \times \sqrt{2} \times \frac{\sqrt{3}+1}{2\sqrt{2}}$$

$$= \frac{(\sqrt{3}+1)}{2} \text{ s.u.}$$

58. We have

$$\cot A + \cot B + \cot C$$

$$= \frac{\cos A}{\sin A} + \frac{\cos B}{\sin B} + \frac{\cos C}{\sin C}$$

$$= \frac{b^2 + c^2 - a^2}{2bc} + \frac{c^2 + a^2 - b^2}{2ac} + \frac{a^2 + b^2 - c^2}{2ab}$$

$$= \frac{b^2 + c^2 - a^2}{2abck} + \frac{c^2 + a^2 - b^2}{2abck} + \frac{a^2 + b^2 - c^2}{2abck}$$

$$= \frac{a^2 + b^2 + c^2}{2abck}$$

$$= \frac{(a^2 + b^2 + c^2)}{4 \times \left(\frac{1}{2}ab\right) \times \sin C}$$

$$= \frac{(a^2 + b^2 + c^2)}{4 \times \Delta} \quad \dots(i)$$

$$\text{Also, } \cot\left(\frac{A}{2}\right) + \cot\left(\frac{B}{2}\right) + \cot\left(\frac{C}{2}\right)$$

$$= \frac{(a+b+c)}{(b+c-a)} \times \cot\left(\frac{A}{2}\right)$$

$$= \frac{s^2}{\Delta} \quad \dots(ii)$$

Dividing (ii) by (i), we get

$$\frac{\cot\left(\frac{A}{2}\right) + \cot\left(\frac{B}{2}\right) + \cot\left(\frac{C}{2}\right)}{\cot A + \cot B + \cot C} = \frac{s^2}{4(a^2 + b^2 + c^2)}$$

$$\Rightarrow \frac{\cot\left(\frac{A}{2}\right) + \cot\left(\frac{B}{2}\right) + \cot\left(\frac{C}{2}\right)}{\cot A + \cot B + \cot C} = \frac{(a+b+c)^2}{(a^2 + b^2 + c^2)}$$

59. Let a, b , and c are the sides of a triangle and s be the semi perimeter.

Let the four quantities are $s, (s-a), (s-b)$ and $(s-c)$

Applying, $AM \geq GM$, we get

$$\Rightarrow \frac{s + (s-a) + (s-b) + (s-c)}{4}$$

$$\geq \sqrt[4]{s(s-a)(s-b)(s-c)}$$

$$\Rightarrow \frac{4s - (a+b+c)}{4} \geq \sqrt[4]{\Delta^2}$$

$$\Rightarrow \frac{4s - 2s}{4} \geq (\Delta)^{\frac{1}{2}}$$

$$\Rightarrow \frac{s}{2} \geq (\Delta)^{\frac{1}{2}}$$

$$\Rightarrow \Delta < \frac{s^2}{4}$$

60. Let $AD = \alpha, BE = \beta$ and $CF = \gamma$

$$\text{Then, } \Delta = \frac{1}{2} \times a \times AD = \frac{1}{2} \times b \times BE = \frac{1}{2} \times c \times CF$$

$$\Rightarrow AD = \frac{2\Delta}{a}, BE = \frac{2\Delta}{b}, CF = \frac{2\Delta}{c}$$

$$\Rightarrow \alpha = \frac{2\Delta}{a}, \beta = \frac{2\Delta}{b}, \gamma = \frac{2\Delta}{c}$$

$$\text{Now, } \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$$

$$= \frac{a^2}{4\Delta^2} + \frac{b^2}{4\Delta^2} + \frac{c^2}{4\Delta^2}$$

$$= \frac{(a^2 + b^2 + c^2)}{4\Delta^2}$$

$$= \frac{1}{\Delta} \times \frac{(a^2 + b^2 + c^2)}{4\Delta}$$

$$= \frac{1}{\Delta} \times (\cot A + \cot B + \cot C)$$

$$= \frac{(\cot A + \cot B + \cot C)}{\Delta}$$

Hence, the result.

61. Let $AD = p_1, BE = p_2$ and $CF = p_3$

$$\text{Then, } \Delta = \frac{1}{2} \times a \times p_1 = \frac{1}{2} \times b \times p_2 = \frac{1}{2} \times c \times p_3$$

$$\Rightarrow p_1 = \frac{2\Delta}{a}, p_2 = \frac{2\Delta}{b}, p_3 = \frac{2\Delta}{c}$$

Now,

$$\begin{aligned}\frac{1}{p_1} + \frac{1}{p_2} - \frac{1}{p_3} &= \frac{a}{2\Delta} + \frac{b}{2\Delta} - \frac{c}{2\Delta} \\ &= \frac{(a+b-c)}{2\Delta} \\ &= \frac{(a+b+c-2c)}{2\Delta} \\ &= \frac{(2s-2c)}{2\Delta} \\ &= \frac{(s-c)}{\Delta} \\ &= \frac{2ab \times s(s-c)}{\Delta \times s} \times \frac{1}{2ab} \\ &= \frac{2ab}{\Delta \times s} \times \frac{s(s-c)}{2ab} \\ &= \frac{2ab}{(a+b+c)\Delta} \times \cos^2\left(\frac{C}{2}\right)\end{aligned}$$

62. We have

$$\begin{aligned}(a-b)^2 + (b-c)^2 + (c-a)^2 &\geq 0 \\ \Rightarrow 2(a^2 + b^2 + c^2) &\geq 2(ab + bc + ca) \\ \Rightarrow 3(a^2 + b^2 + c^2) &\geq (a^2 + b^2 + c^2) + (2ab + bc + ca) \\ \Rightarrow 3(a^2 + b^2 + c^2) &> (a+b+c)^2 > d^2 \\ \Rightarrow \frac{3(a^2 + b^2 + c^2)}{d^2} &> 1 \\ \Rightarrow \frac{(a^2 + b^2 + c^2)}{d^2} &> \frac{1}{3}\end{aligned}$$

Thus, the minimum value of $\frac{(a^2 + b^2 + c^2)}{d^2}$ is $\frac{1}{3}$

63. We have

$$\begin{aligned}\cos A + \cos B + \cos C &= \frac{3}{2} \\ \Rightarrow 1 + 4 \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right) &= \frac{3}{2} \\ \Rightarrow \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right) &= \frac{1}{8}\end{aligned}$$

It is possible only when

$$\begin{aligned}\sin\left(\frac{A}{2}\right) &= \frac{1}{2}, \sin\left(\frac{B}{2}\right) = \frac{1}{2}, \sin\left(\frac{C}{2}\right) = \frac{1}{2} \\ \Rightarrow \frac{A}{2} &= \frac{\pi}{6}, \frac{B}{2} = \frac{\pi}{6}, \frac{C}{2} = \frac{\pi}{6} \\ \Rightarrow A &= \frac{\pi}{3}, B = \frac{\pi}{3}, C = \frac{\pi}{3} \\ \Rightarrow \Delta &\text{ is an equilateral.}\end{aligned}$$

64. Here,

$$\begin{aligned}\Delta &= \frac{1}{2} \times p \times p_1 = \frac{1}{2} \times q \times p_2 = \frac{1}{2} \times r \times p_3 \\ \Rightarrow p &= \frac{2\Delta}{p_1}, q = \frac{2\Delta}{p_2}, r = \frac{2\Delta}{p_3}\end{aligned}$$

From sine rule of a triangle,

$$\frac{\sin P}{p} = \frac{\sin Q}{q} = \frac{\sin R}{r}$$

Given $\sin P, \sin Q, \sin R$ are in AP

$$\Rightarrow p, q, r \in \text{AP}$$

$$\Rightarrow \frac{2\Delta}{p_1}, \frac{2\Delta}{p_2}, \frac{2\Delta}{p_3} \in \text{AP}$$

$$65. \text{ In } \Delta ABC, \Delta = \frac{1}{2} \times ac \sin(\angle B)$$

$$\Rightarrow (6 + 2\sqrt{3}) = \frac{1}{2} \times 2(\sqrt{3} + 1) \times c \times \frac{1}{\sqrt{2}}$$

$$\Rightarrow c = \frac{\sqrt{2}(6 + 2\sqrt{3})}{(\sqrt{3} + 1)}$$

$$\Rightarrow c = \frac{2\sqrt{3}\sqrt{2}(\sqrt{3} + 1)}{(\sqrt{3} + 1)}$$

$$\Rightarrow c = 2\sqrt{6}$$

$$\text{Now, } \cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \frac{4(\sqrt{3} + 1)^2 + 24 - b^2}{2 \cdot 2(\sqrt{3} + 1) \cdot 2\sqrt{6}}$$

$$\Rightarrow 4(\sqrt{3} + 1)^2 + 24 - b^2 = 8\sqrt{3}(\sqrt{3} + 1)$$

$$\Rightarrow 4(4 + 2\sqrt{3}) + 24 - b^2 = 8(3 + \sqrt{3})$$

$$\Rightarrow b^2 = 16$$

$$\Rightarrow b = 4$$

$$66. \text{ Let } \angle B = 30^\circ, \angle C = 45^\circ$$

$$\text{So, } \angle A = 180^\circ - (30^\circ + 45^\circ) = 105^\circ$$

From sine formula, we can write

$$\frac{a}{\sin(105^\circ)} = \frac{b}{\sin(30^\circ)} = \frac{c}{\sin(45^\circ)}$$

$$\frac{(\sqrt{3} + 1)}{\left(\frac{\sqrt{3} + 1}{2\sqrt{2}}\right)} = \frac{b}{\frac{1}{2}} = \frac{c}{\frac{1}{\sqrt{2}}}$$

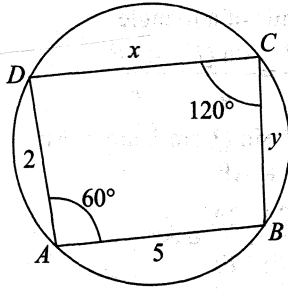
$$2\sqrt{2} = 2b = c\sqrt{2}$$

$$b = \sqrt{2}, c = 2$$

Thus, area of triangle ABC is

$$\begin{aligned}&= \frac{1}{2} bc \sin A \\ &= \frac{1}{2} \times 2 \times \sqrt{2} \times \sin(105^\circ) \\ &= \frac{1}{2} \times 2 \times \sqrt{2} \times \left(\frac{\sqrt{3} + 1}{2\sqrt{2}}\right) \\ &= \left(\frac{\sqrt{3} + 1}{2}\right) \text{ s.u.}\end{aligned}$$

$$67. \text{ Suppose, } AC = 2, AB = 5, BC = x, CD = y \text{ and } \angle BAD = 60^\circ$$



$$\begin{aligned}\text{Area of } \triangle ABC &= \frac{1}{2} \cdot 5 \cdot 2 \cdot \sin(60^\circ) \\ &= \frac{5\sqrt{3}}{2}\end{aligned}$$

Also, from $\triangle ABC$,

$$\begin{aligned}\cos(60^\circ) &= \frac{25 + 4 - BD^2}{2 \cdot 5 \cdot 2} \\ &= \frac{29 - BD^2}{20}\end{aligned}$$

$$\Rightarrow \frac{29 - BD^2}{20} = \frac{1}{2}$$

$$\Rightarrow BD^2 = 19$$

$$\Rightarrow BD = \sqrt{19}$$

Since A, B, C, D are concyclic, so
 $\angle BCD = 180^\circ - 60^\circ = 120^\circ$

Then, from $\triangle BCD$,

$$\cos(120^\circ) = \frac{x^2 + y^2 - (\sqrt{19})^2}{2xy}$$

$$\Rightarrow \frac{x^2 + y^2 - (\sqrt{19})^2}{2xy} = -\frac{1}{2}$$

$$\Rightarrow \frac{x^2 + y^2 - (\sqrt{19})^2}{xy} = -1$$

$$\Rightarrow x^2 + y^2 + xy = 19$$

Again, area of $\triangle BCD$

$$= \frac{1}{2} \cdot x \cdot y \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}xy}{4}$$

Thus, area of quad. $ABCD = 4\sqrt{3}$

$$\Rightarrow \frac{5\sqrt{3}}{2} + \frac{\sqrt{3}xy}{4} = 4\sqrt{3}$$

$$\Rightarrow \frac{5}{2} + \frac{xy}{4} = 4$$

$$\Rightarrow \frac{xy}{4} = 4 - \frac{5}{2} = \frac{3}{2}$$

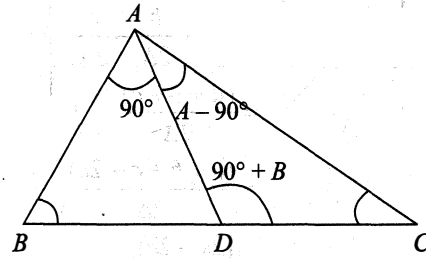
$$\Rightarrow xy = 6$$

From (i), we get

$$x^2 + y^2 = 13$$

$$\Rightarrow x = 3, y = 2$$

68. Since AD is the median, so $BD : DC = 1 : 1$



Clearly, $\angle ADC = 90^\circ + B$.

Now, applying $m : n$ rule, we get,

$$(1 + 1) \cot(90^\circ + B) = 1 \cdot \cot(90^\circ) - 1 \cdot \cot(A - 90^\circ)$$

$$\Rightarrow -2 \tan B = 0 - (-\tan A)$$

$$\Rightarrow -2 \tan B = \tan A$$

$$\Rightarrow \tan A + 2 \tan B = 0$$

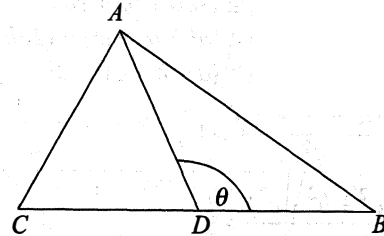
Hence, the result.

$$\Rightarrow \frac{1}{p_1}, \frac{1}{p_2}, \frac{1}{p_3} \in AP$$

$$\Rightarrow p_1, p_2, p_3 \in HP$$

Thus, the altitudes are in HP

69.



By $m : n$ rule, we get

$$(1 + 1) \cot \theta = 1 \cdot \cot C - 1 \cdot \cot B$$

$$\Rightarrow 2 \cot \theta = \cot C - \cot B$$

$$\Rightarrow 2 \cot \theta = \frac{a^2 + b^2 - c^2}{2ab \sin C} - \frac{a^2 + c^2 - b^2}{2ab \sin B}$$

$$\Rightarrow 2 \cot \theta = \frac{a^2 + b^2 - c^2}{4\Delta} - \frac{a^2 + c^2 - b^2}{4\Delta}$$

$$\Rightarrow 2 \cot \theta = \frac{2(b^2 - c^2)}{4\Delta}$$

$$\Rightarrow \cot \theta = \frac{(b^2 - c^2)}{4\Delta}$$

Hence, the result.

70. Clearly, the triangle is right angled.

$$(\because 18^2 + 24^2 = 30^2)$$

Thus, the area of the triangle

$$= \frac{1}{2} \times 24 \times 18 = 12 \times 18$$

Therefore, the circum-radius

$$= R$$

$$= \frac{abc}{4\Delta}$$

$$= \frac{18 \times 24 \times 30}{4 \times 12 \times 18} = 15$$

71. As we know that,

$$\frac{a}{\sin A} = 2R$$

$$\Rightarrow 2R = \frac{2\sqrt{3}}{\sin(60^\circ)}$$

$$\Rightarrow R = \frac{\sqrt{3}}{\frac{\sqrt{3}}{2}} = 2$$

Hence, the circum-radius is 2.

72. Let $a = 3$, $b = 4$ and $c = 5$

Clearly, it is a right angled triangle

$$\text{Thus, } \Delta = \frac{1}{2} \times 4 \times 3 = 6 \text{ sq.u}$$

Hence, the circum-radius R

$$\begin{aligned} &= \frac{abc}{4\Delta} \\ &= \frac{3 \times 4 \times 5}{6} \\ &= 10 \end{aligned}$$

73. We have

$$8R^2 = a^2 + b^2 + c^2 \quad 8R^2 = (2R \sin A)^2 + (2R \sin B)^2 + (2R \sin C)^2$$

$$\Rightarrow \sin^2 A + \sin^2 B + \sin^2 C = 2$$

$$\Rightarrow 1 - \cos^2 A + 1 - \cos^2 B + \sin^2 C = 2$$

$$\Rightarrow \cos^2 A - \sin^2 C + \cos^2 B = 0$$

$$\Rightarrow \cos(A+C) \cos(A-C) + \cos^2 B = 0$$

$$\Rightarrow \cos(\pi - B) \cos(A-C) + \cos^2 B = 0$$

$$\Rightarrow \cos B \cos(A-C) - \cos^2 B = 0$$

$$\Rightarrow \cos B (\cos(A-C) - \cos B) = 0$$

$$\Rightarrow \cos B (\cos(A-C) + \cos(A+C)) = 0$$

$$\Rightarrow \cos B \cdot 2 \cos A \cos C = 0$$

$$\Rightarrow \cos A = 0, \cos B = 0, \cos C = 0$$

$$\Rightarrow A = \frac{\pi}{2} \text{ or } B = \frac{\pi}{2} \text{ or } C = \frac{\pi}{2}$$

Thus, the triangle is right angled.

74. We have $a \cos A + b \cos B + c \cos C$

$$= 2R(\sin A \cdot \cos A + \sin B \cdot \cos B + \sin C \cdot \cos C)$$

$$= \frac{2R}{2} [2 \sin A \cdot \cos A + 2 \sin B \cdot \cos B + 2 \sin C \cdot \cos C]$$

$$= R(\sin 2A + \sin 2B + \sin 2C)$$

$$= R(4 \sin A \cdot \sin B \cdot \sin C)$$

$$= 4R \sin A \cdot \sin B \cdot \sin C$$

75. We have

$$\Delta = \frac{1}{2} \times a \times b \times \sin C$$

$$\Rightarrow \Delta = \frac{1}{2} \times 2R \sin A \times 2R \sin B \times \sin C$$

$$\Rightarrow \Delta = 2R^2 \cdot \sin A \cdot \sin B \cdot \sin C$$

76. We have

$$\begin{aligned} &\frac{\sin A}{a} + \frac{\sin B}{b} + \frac{\sin C}{c} \\ &= \frac{\sin A}{2R \sin A} + \frac{\sin B}{2R \sin B} + \frac{\sin C}{2R \sin C} \\ &= \frac{1}{2R} + \frac{1}{2R} + \frac{1}{2R} \\ &= \frac{3}{2R} \end{aligned}$$

77. Let $AD = p_1$, $BE = p_2$ and $CF = p_3$.

Then,

$$\Delta = \frac{1}{2} \times a \times p_1 = \frac{1}{2} \times b \times p_2 = \frac{1}{2} \times c \times p_3$$

$$\Rightarrow p_1 = \frac{2\Delta}{a}, p_2 = \frac{2\Delta}{b}, p_3 = \frac{2\Delta}{c}$$

Now,

$$\begin{aligned} \frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} &= \frac{a+b+c}{2\Delta} \\ &= \frac{2R(\sin A + \sin B + \sin C)}{2\Delta} \\ &\leq \frac{3R}{\Delta} \quad (\because \sin A \leq 1, \sin B \leq 1, \sin C \leq 1) \end{aligned}$$

78. Here, $\Delta = \frac{1}{2}ap_1 = \frac{1}{2}bp_2 = \frac{1}{2}cp_3$

$$\Rightarrow p_1 = \frac{2\Delta}{a}, p_2 = \frac{2\Delta}{b}, p_3 = \frac{2\Delta}{c}$$

Now,

$$\begin{aligned} \frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} &= \frac{a+b+c}{2\Delta} \\ &= \frac{2s}{2\Delta} \\ &= \frac{s}{\Delta} \\ &= \frac{1}{r} \end{aligned}$$

79. We have

$$\begin{aligned} &\frac{\cos A}{p_1} + \frac{\cos B}{p_2} + \frac{\cos C}{p_3} \\ &= \frac{1}{2\Delta} (a \cos A + b \cos B + c \cos C) \\ &= \frac{1}{2\Delta} [2R \sin A \cos A + 2R \sin B \cos B + 2R \sin C \cos C] \end{aligned}$$

$$\begin{aligned}
 &= \frac{R}{2\Delta} (\sin 2A + \sin 2B + \sin 2C) \\
 &= \frac{R}{2\Delta} (4 \sin A \sin B \sin C) \\
 &= \frac{2R}{\Delta} (\sin A \sin B \sin C) \\
 &= \frac{2R}{\Delta} \times \left(\frac{a}{2R}\right) \times \left(\frac{b}{2R}\right) \times (\sin C) \\
 &= \frac{1}{\Delta R} \left(\frac{1}{2} ab \sin C\right) \\
 &= \frac{1}{\Delta R} \times \Delta \\
 &= \frac{1}{R}
 \end{aligned}$$

80. We have

$$\begin{aligned}
 \frac{\cos C}{\sqrt{4R^2 - c^2}} &= \frac{\cos C}{\sqrt{4R^2 - 4R^2 \sin^2 C}} \\
 &= \frac{\cos C}{\sqrt{4R^2 (1 - \sin^2 C)}} \\
 &= \frac{\cos C}{2R \cos C} = \frac{1}{2R}
 \end{aligned}$$

81. We have

$$\begin{aligned}
 \Delta &= \frac{1}{2} \times p_1 \times a \\
 p_1 &= \frac{2\Delta}{a}
 \end{aligned}$$

Similarly,

$$p_2 = \frac{2\Delta}{b}, p_3 = \frac{2\Delta}{c}$$

Now,

$$\begin{aligned}
 p_1 p_2 p_3 &= \frac{8\Delta^3}{abc} \\
 &= \frac{8 \left(\frac{abc}{4R}\right)^3}{abc} \\
 &= \frac{a^2 b^2 c^2}{8R^3}
 \end{aligned}$$

82. We have

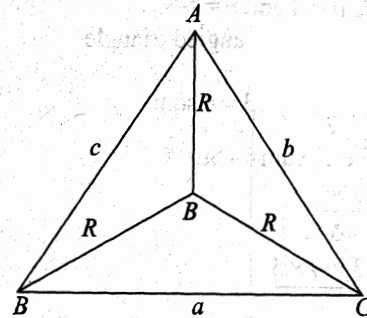
$$\begin{aligned}
 \Delta &= \frac{1}{2} \times p_1 \times a \\
 p_1 &= \frac{2\Delta}{a}
 \end{aligned}$$

Similarly, $p_2 = \frac{2\Delta}{b}, p_3 = \frac{2\Delta}{c}$

$$\text{Now, } \frac{bp_1}{c} + \frac{cp_2}{a} + \frac{ap_3}{b}$$

$$\begin{aligned}
 &= \frac{2\Delta b}{ac} + \frac{2\Delta c}{ab} + \frac{2\Delta a}{bc} \\
 &= 2\Delta \left(\frac{b}{ac} + \frac{c}{ab} + \frac{a}{bc}\right) \\
 &= \frac{2\Delta(b^2 + c^2 + a^2)}{abc} \\
 &= \frac{(a^2 + b^2 + c^2)}{2R}
 \end{aligned}$$

83. Given, O is the circumcentre of ΔABC



Let, $\text{ar}(\Delta BOC) = \Delta_1$, $\text{ar}(\Delta AOC) = \Delta_2$ and $\text{ar}(\Delta AOB) = \Delta_3$ respectively.

$$\text{Now, } R_1 = \frac{OB \cdot OC \cdot a}{4(\Delta BOC)} = \frac{aR^2}{\Delta_1}$$

$$\frac{a}{R_1} = \frac{4\Delta_1}{R^2}$$

$$\text{Similarly, } \frac{b}{R_2} = \frac{4\Delta_2}{R^2}, \frac{c}{R_3} = \frac{4\Delta_3}{R^2}$$

$$\text{Now, } \frac{a}{R_1} + \frac{b}{R_2} + \frac{c}{R_3}$$

$$= \frac{4}{R^2} (\Delta_1 + \Delta_2 + \Delta_3)$$

$$= \frac{4\Delta}{R^2}$$

$$= \frac{4}{R^2} \times \frac{abc}{4R} = \frac{abc}{R^3}$$

$$\frac{a \sec A + b \sec B + c \sec C}{\tan A \tan B \tan C} = 2$$

84. We have

$$\frac{a \sec A + b \sec B + c \sec C}{\tan A \tan B \tan C}$$

$$= \frac{2R \sin A \sec A + 2R \sin B \sec B + 2R \sin C \sec C}{\tan A \tan B \tan C}$$

$$= \frac{2R \tan A + 2R \tan B + 2R \tan C}{\tan A \tan B \tan C}$$

$$= \frac{2R (\tan A + \tan B + \tan C)}{\tan A \tan B \tan C}$$

$$= \frac{2R(\tan A \cdot \tan B \cdot \tan C)}{\tan A \tan B \tan C}$$

$$= 2R$$

85. We have

$$\begin{aligned} a \cos A + b \cos B + c \cos C &= 2R(\sin A \cos A + \sin B \cos B + \sin C \cos C) \\ &= R(2 \sin A \cos A + 2 \sin B \cos B + 2 \sin C \cos C) \\ &= R(\sin 2A + \sin 2B + \sin 2C) \\ &= R(4 \sin A \cdot \sin B \cdot \sin C) \\ &= 4(R \sin A \cdot \sin B \cdot \sin C) \end{aligned}$$

86. Here, $s = \frac{4+6+8}{2} = 9$

Area of a triangle

$$\begin{aligned} &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{9(9-4)(9-6)(9-8)} \\ &= \sqrt{9 \times 5 \times 3 \times 1} = 3\sqrt{15} \end{aligned}$$

Hence, the in-radius

$$= r = \frac{\Delta}{s} = \frac{3\sqrt{15}}{9} = \frac{\sqrt{15}}{3}$$

87. Clearly, it is a right angled triangle

So, its area = $\frac{1}{2} \times 18 \times 24 = 9 \times 24$

and $s = \frac{18+24+30}{2} = \frac{72}{2} = 36$

Thus, in-radius = $r = \frac{\Delta}{s} = \frac{9 \times 24}{36} = 6$

88. Do yourself.

89. Two sides of a triangle are 2 and $\sqrt{3}$ and the included angle is 30° , then prove that its in-radius is $\frac{1}{2}(\sqrt{3} - 1)$.

89. We have area of the triangle

$$\begin{aligned} \Delta &= \frac{1}{2} \times 2 \times \sqrt{3} \times \sin(30^\circ) \\ &= \frac{\sqrt{3}}{2} \end{aligned}$$

Also, $a^2 = b^2 + c^2 - 2bc \cos A$

$$a^2 = 4 + 3 - 2 \times 2 \times \sqrt{3} \times \cos(30^\circ)$$

$$a^2 = 7 - 2 \times 2 \times \sqrt{3} \times \frac{\sqrt{3}}{2}$$

$$a^2 = 7 - 6 = 1$$

$$a = 1$$

Now, $s = \frac{a+b+c}{2} = \frac{1+2+\sqrt{3}}{2} = \frac{3+\sqrt{3}}{2}$

Hence, in-radius

$$\begin{aligned} &= r = \frac{\Delta}{s} \\ &= \frac{\frac{\sqrt{3}}{2}}{\frac{3+\sqrt{3}}{2}} = \frac{\sqrt{3}}{(\sqrt{3}+3)} \\ &= \frac{1}{(\sqrt{3}+1)} = \frac{(\sqrt{3}-1)}{2} \end{aligned}$$

90. We have

$$\begin{aligned} \frac{R}{r} &= \frac{\frac{abc}{4\Delta}}{\frac{\Delta}{s}} = \frac{a^3}{4\Delta} \times \frac{s}{\Delta} = \frac{a^3}{4\Delta} \times \frac{3a}{2\Delta} \\ &= \frac{3a^4}{8} \times \frac{1}{\Delta^2} \\ &= \frac{3a^4}{8} \times \frac{1}{\frac{3a^4}{8}} = \frac{3a^4}{8} \times \frac{16}{3a^4} = 2 \end{aligned}$$

91. We have

$$\begin{aligned} \frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} &= \frac{a+b+c}{abc} \\ &= \frac{2s}{4\Delta R} \\ &= \frac{\Delta}{2\Delta R} \\ &= \frac{1}{2rR} \end{aligned}$$

92. We have

$$\cos A + \cos B + \cos C$$

$$\begin{aligned} &= 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right) + \cos C \\ &= 2 \cos\left(\frac{\pi}{2} - \frac{C}{2}\right) \cos\left(\frac{A-B}{2}\right) + \cos C \\ &= 2 \sin\left(\frac{C}{2}\right) \cos\left(\frac{A-B}{2}\right) + \cos C \\ &= 2 \sin\left(\frac{C}{2}\right) \cos\left(\frac{A-B}{2}\right) + 1 - 2 \sin^2\left(\frac{C}{2}\right) \\ &= 1 + 2 \sin\left(\frac{C}{2}\right) \left(\cos\left(\frac{A-B}{2}\right) - \sin^2\left(\frac{C}{2}\right) \right) \\ &= 1 + 2 \sin\left(\frac{C}{2}\right) \left(\cos\left(\frac{A-B}{2}\right) - \cos\left(\frac{A+B}{2}\right) \right) \\ &= 1 + 2 \sin\left(\frac{C}{2}\right) \left(2 \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \right) \\ &= 1 + 4 \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right) \\ &= \left(1 + \frac{r}{R}\right) \end{aligned}$$

93. We have,

$$\begin{aligned}\sin A + \sin B + \sin C &= \frac{a}{2R} + \frac{b}{2R} + \frac{c}{2R} \\ &= \frac{a+b+c}{2R} \\ &= \frac{2s}{2R} \\ &= \frac{s}{R} \\ &= \frac{\Delta}{R} \\ &= \frac{r}{R} \\ &= \frac{\Delta}{rR}\end{aligned}$$

94. We have

$$\begin{aligned}a \cot A + b \cot B + c \cot C &= \left[2R \sin A \times \frac{\cos A}{\sin A} + 2R \sin B \times \frac{\cos B}{\sin B} \right. \\ &\quad \left. + 2R \sin C \times \frac{\cos C}{\sin C} \right] \\ &= 2R(\cos A + \cos B + \cos C) \\ &= 2R \left(1 + \frac{r}{R} \right) \\ &= 2(R+r)\end{aligned}$$

95. We have $r_1 = R$

$$\begin{aligned}4R \sin\left(\frac{A}{2}\right) \cos\left(\frac{B}{2}\right) \cos\left(\frac{C}{2}\right) &= R \\ 2 \sin\left(\frac{A}{2}\right) \left\{ 2 \cos\left(\frac{B}{2}\right) \cos\left(\frac{C}{2}\right) \right\} &= 1 \\ 2 \sin\left(\frac{A}{2}\right) \left\{ \cos\left(\frac{B+C}{2}\right) + \cos\left(\frac{B-C}{2}\right) \right\} &= 1 \\ 2 \cos\left(\frac{B+C}{2}\right) \left\{ \cos\left(\frac{B+C}{2}\right) + \cos\left(\frac{B-C}{2}\right) \right\} &= 1 \\ 2 \cos^2\left(\frac{B+C}{2}\right) + \left\{ 2 \cos\left(\frac{B+C}{2}\right) \cos\left(\frac{B-C}{2}\right) \right\} &= 1\end{aligned}$$

$$\begin{aligned}(1 + \cos(B+C)) &= \cos(B) + \cos(C) = 1 \\ \cos(B) + \cos(C) &= -\cos(B+C) = \cos A \\ \text{Hence, the result.}\end{aligned}$$

96. We have

$$\begin{aligned}(b+c) \tan\left(\frac{A}{2}\right) &= 2R(\sin B + \sin C) \tan\left(\frac{A}{2}\right) \\ &= 2R \times 2 \sin\left(\frac{B+C}{2}\right) \cos\left(\frac{B-C}{2}\right) \times \tan\left(\frac{A}{2}\right) \\ &= 2R \times 2 \cos\left(\frac{A}{2}\right) \cos\left(\frac{B-C}{2}\right) \times \frac{\sin\left(\frac{A}{2}\right)}{\cos\left(\frac{A}{2}\right)}\end{aligned}$$

$$\begin{aligned}&= 2R \times 2 \cos\left(\frac{B-C}{2}\right) \times \sin\left(\frac{A}{2}\right) \\ &= 2R \times 2 \cos\left(\frac{B-C}{2}\right) \times \cos\left(\frac{B+C}{2}\right) \\ &= 2R \times (\cos(B) + \cos(C))\end{aligned}$$

Thus, LHS

$$\begin{aligned}&= 4R(\cos A + \cos B + \cos C) \\ &= 4R \left(1 + \frac{r}{R} \right) \\ &= 4(R+r)\end{aligned}$$

97. We have

$$\begin{aligned}R &= \frac{c}{2 \sin C} = \frac{c}{2 \sin(90^\circ)} = \frac{c}{2} \\ c &= 2R\end{aligned}$$

$$\text{Also, } r = (s-c) \tan\left(\frac{C}{2}\right)$$

$$\begin{aligned}r &= (s-c) \tan(45^\circ) = (s-c) \\ 2r &= (2s-2c) = (a+b+c-2c) \\ 2r &= (a+b-c) \\ 2r &= (a+b-2R) \\ 2(R+r) &= (a+b)\end{aligned}$$

98. We have

$$\begin{aligned}\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2} + \cos^2 \frac{C}{2} &= \frac{1}{2} \left(2 \cos^2 \left(\frac{A}{2} \right) + 2 \cos^2 \left(\frac{B}{2} \right) + 2 \cos^2 \left(\frac{C}{2} \right) \right) \\ &= \frac{1}{2} (1 + \cos A + 1 + \cos B + 1 + \cos C) \\ &= \frac{1}{2} (3 + \cos A + \cos B + \cos C) \\ &= \frac{1}{2} \left(3 + \left(1 + \frac{r}{R} \right) \right) \\ &= \frac{1}{2} \left(4 + \frac{r}{R} \right) \\ &= \left(2 + \frac{r}{2R} \right)\end{aligned}$$

99. Let O is the circum-centre and $OD = x$, $OE = y$, $OF = z$ respectively.

Also, $OA = R = OB = OC$

We have $x = OD = R \cos A$

$$\Rightarrow \frac{a}{2 \sin A} \cdot \cos A = \frac{a}{2 \tan A}$$

$$\tan A = \frac{a}{2x}$$

$$\text{Similarly, } \tan B = \frac{b}{2y} \text{ and } \tan C = \frac{c}{2z}$$

As we know that, in a $\triangle ABC$,

$$\tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$$

$$\Rightarrow \frac{a}{2x} + \frac{b}{2y} + \frac{c}{2z} = \frac{a}{2x} \cdot \frac{b}{2y} \cdot \frac{c}{2z}$$

$$\Rightarrow \frac{a}{x} + \frac{b}{y} + \frac{c}{z} = \frac{a \cdot b \cdot c}{4 \cdot x \cdot y \cdot z}$$

100. We have

$$R_1 = \frac{OB \cdot OC \cdot BC}{4\Delta_{OBC}} = \frac{R \cdot R \cdot a}{4\Delta_1} = \frac{R^2 \cdot a}{4\Delta_1}$$

$$\Rightarrow \frac{a}{R_1} = \frac{4\Delta_1}{R^2}$$

$$\text{Similarly, } \frac{b}{R_2} = \frac{4\Delta_2}{R^2} \text{ and } \frac{c}{R_3} = \frac{4\Delta_3}{R^2}$$

Thus,

$$\begin{aligned} \frac{a}{R_1} + \frac{b}{R_2} + \frac{c}{R_3} &= \frac{4\Delta_1}{R^2} + \frac{4\Delta_2}{R^2} + \frac{4\Delta_3}{R^2} \\ &= \frac{4(\Delta_1 + \Delta_2 + \Delta_3)}{R^2} \\ &= \frac{4\Delta}{R^2} \\ &= \frac{4\Delta}{R^2} \end{aligned}$$

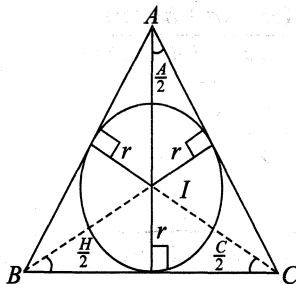
101. Let $AD = p_1$, $BE = p_2$ and $CF = p_3$

$$\text{Then, } \Delta = \frac{1}{2}ap_1 = \frac{1}{2}bp_2 = \frac{1}{2}cp_3$$

$$\Rightarrow p_1 = \frac{2\Delta}{a}, p_2 = \frac{2\Delta}{b}, p_3 = \frac{2\Delta}{c}$$

$$\begin{aligned} \text{We have } p_1 \cdot p_2 \cdot p_3 &= \frac{2\Delta}{a} \cdot \frac{2\Delta}{b} \cdot \frac{2\Delta}{c} \\ &= \frac{8\Delta^3}{abc} \\ &= \frac{8\left(\frac{abc}{4R}\right)^3}{abc} \end{aligned}$$

102.



Since IA is the internal angle bisector of $\angle A$, so we can write

$$\begin{aligned} \frac{AB}{AC} &= \frac{BD}{DC} \\ \Rightarrow \frac{BD}{DC} &= \frac{AB}{AC} = \frac{c}{b} \end{aligned}$$

$$\Rightarrow \frac{DC}{BD} + 1 = \frac{b}{c} + 1$$

$$\Rightarrow \frac{BC}{BD} = \frac{b+c}{c}$$

$$\Rightarrow BD = \frac{ac}{b+c}$$

$$\text{In } \triangle ABD, \frac{BD}{\sin\left(\frac{A}{2}\right)} = \frac{AD}{\sin B}$$

$$\begin{aligned} \Rightarrow AD &= \frac{BD \sin B}{\sin\left(\frac{A}{2}\right)} \\ &= \frac{ac \sin B}{(b+c) \sin\left(\frac{A}{2}\right)} = \frac{2\Delta}{(b+c) \sin(A/2)} \end{aligned}$$

$$\text{similarly, } BE = \frac{2\Delta}{(c+a) \sin\left(\frac{B}{2}\right)}$$

$$\text{and } CF = \frac{2\Delta}{(a+b) \sin\left(\frac{C}{2}\right)}$$

103. Now $2s = a + b + c = 18 + 24 + 30 = 72$

$$\Rightarrow 2s = 72$$

$$\Rightarrow s = 36$$

$$\begin{aligned} \text{We have } \Delta &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{36(36-18)(36-24)(36-30)} \\ &= \sqrt{36 \times 18 \times 12 \times 6} \\ &= \sqrt{36 \times 9 \times 12 \times 12} \\ &= 6 \times 3 \times 12 \\ &= 216 \end{aligned}$$

$$\text{Thus, } r_1 = \frac{\Delta}{s-a} = \frac{216}{36-18} = \frac{216}{18} = 12$$

$$r_2 = \frac{\Delta}{s-b} = \frac{216}{36-24} = \frac{216}{12} = 18$$

$$\text{and } r_3 = \frac{\Delta}{s-c} = \frac{216}{36-30} = \frac{216}{6} = 36$$

$$\begin{aligned} 104. \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} &= \frac{s-a}{\Delta} + \frac{s-b}{\Delta} + \frac{s-c}{\Delta} \\ &= \frac{3s - (a+b+c)}{\Delta} \\ &= \frac{3s - 2s}{\Delta} \\ &= \frac{s}{\Delta} \\ &= \frac{1}{r} \end{aligned}$$

$$\begin{aligned}
 105. \quad & \frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3} \\
 &= \frac{(b-c)(s-a)}{\Delta} + \frac{(c-a)(s-b)}{\Delta} + \frac{(a-b)(s-c)}{\Delta} \\
 &= \frac{1}{\Delta} [s(b-c+c-a+a-b)] \\
 &\quad - \frac{1}{\Delta} (ab-ac+bc-ab+ca-bc) \\
 &= 0-0 \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 106. \quad & \text{Given, } \frac{s-c}{s-a} = \frac{b-c}{a-b} \\
 \Rightarrow & \frac{2s-2c}{2s-2a} = \frac{b-c}{a-b} \\
 \Rightarrow & \frac{a+b-c}{b+c-a} = \frac{b-c}{a-b} \\
 \Rightarrow & \frac{a+b-c}{b-c} = \frac{b+c-a}{a-b} \\
 \Rightarrow & \frac{a}{b-c} + 1 = \frac{c}{a-b} - 1 \\
 \Rightarrow & \frac{c}{a-b} + \frac{a}{b-c} = 2 \\
 \Rightarrow & \frac{c(b-c) + a(a-b)}{(a-b)(b-c)} = 2 \\
 \Rightarrow & \frac{bc-c^2+a^2-ab}{(a-b)(b-c)} = 2 \\
 \Rightarrow & bc-c^2+a^2-ab = 2(ab-ac-b^2+bc) \\
 \Rightarrow & 2b^2-b-c^2+a^2-3ab+2ac=0
 \end{aligned}$$

107. We have

$$\begin{aligned}
 & \left(1 - \frac{r_1}{r_2}\right) \left(1 - \frac{r_1}{r_3}\right) = 2 \\
 & \left(1 - \frac{s-b}{s-a}\right) \left(1 - \frac{s-c}{s-a}\right) = 2 \\
 & \left(\frac{s-a-s+b}{s-a}\right) \left(\frac{s-a-s+c}{s-a}\right) = 2 \\
 & \left(\frac{a-b}{s-a}\right) \left(\frac{a-c}{s-a}\right) = 2 \\
 & \frac{(a-b)(a-c)}{(s-a)^2} = 2 \\
 & (a-b)(a-c) = 2(s-a)^2 \\
 & 2(a-b)(a-c) = 4(s-a)^2 = (2s-2a)^2 \\
 & 2(a-b)(a-c) = (b+c-a)^2 \\
 & 2(a^2-ab-ac+bc) \\
 & \quad = a^2+b^2+c^2+2bc-2ab-2ac \\
 & a^2+b^2=c^2
 \end{aligned}$$

Thus, the $\triangle ABC$ is a right angled.

108. We have

$$\begin{aligned}
 & r_1 + r_2 + r_3 - r \\
 &= \left\{ \frac{\Delta}{(s-a)} + \frac{\Delta}{(s-b)} \right\} + \left\{ \frac{\Delta}{(s-c)} - \frac{\Delta}{s} \right\} \\
 &= \left\{ \frac{\Delta(s-b+s-a)}{(s-a)(s-b)} \right\} + \left\{ \frac{\Delta(s-s+c)}{s(s-c)} \right\} \\
 &= \left\{ \frac{\Delta(2s-b-a)}{(s-a)(s-b)} \right\} + \left\{ \frac{\Delta(c)}{s(s-c)} \right\} \\
 &= \left\{ \frac{\Delta(a+b+c-b-a)}{(s-a)(s-b)} \right\} + \left\{ \frac{\Delta(c)}{s(s-c)} \right\} \\
 &= \left\{ \frac{\Delta(c)}{(s-a)(s-b)} \right\} + \left\{ \frac{\Delta(c)}{s(s-c)} \right\} \\
 &= \Delta(c) \left\{ \frac{1}{(s-a)(s-b)} + \frac{1}{s(s-c)} \right\} \\
 &= \Delta(c) \left\{ \frac{s(s-c) + (s-a)(s-b)}{s(s-a)(s-b)(s-c)} \right\} \\
 &= \Delta(c) \left\{ \frac{s^2 - cs + s^2 - (a+b)s + ab}{\Delta^2} \right\} \\
 &= c \times \left\{ \frac{2s^2 - (a+b+c)s + ab}{\Delta} \right\} \\
 &= c \times \left\{ \frac{2s^2 - 2s^2 + ab}{\Delta} \right\} \\
 &= \frac{abc}{\Delta} = 4R
 \end{aligned}$$

109. $r_1 r_2 + r_2 r_3 + r_3 r_1$

$$\begin{aligned}
 &= \frac{\Delta}{(s-a)} \cdot \frac{\Delta}{(s-b)} + \frac{\Delta}{(s-b)} \cdot \frac{\Delta}{(s-c)} \\
 &\quad + \frac{\Delta}{(s-c)} \cdot \frac{\Delta}{(s-a)} \\
 &= \frac{\Delta^2(s-c+s-a+s-b)}{(s-a)(s-b)(s-c)} \\
 &= \frac{\Delta^2(3s-(a+b+c))}{(s-a)(s-b)(s-c)} \\
 &= \frac{\Delta^2(3s-2s)}{(s-a)(s-b)(s-c)} \\
 &= \frac{\Delta^2 \times s}{(s-a)(s-b)(s-c)} \\
 &= \frac{\Delta^2 \times s^2}{s(s-a)(s-b)(s-c)} \\
 &= \frac{\Delta^2 \times s^2}{\Delta^2} \\
 &= s^2
 \end{aligned}$$

$$\begin{aligned}
 110. \quad & r_1 + r_2 - r_3 + r \\
 &= (r_1 + r_2) - (r_3 - r) \\
 &= \left(\frac{\Delta}{(s-a)} + \frac{\Delta}{(s-b)} \right) - \left(\frac{\Delta}{(s-c)} - \frac{\Delta}{s} \right) \\
 &= \Delta \left\{ \left(\frac{1}{(s-a)} + \frac{1}{(s-b)} \right) - \left(\frac{1}{(s-c)} - \frac{1}{s} \right) \right\} \\
 &= \Delta \left\{ \left(\frac{(s-a+s-b)}{(s-a)(s-b)} \right) - \left(\frac{s-(s-c)}{s(s-c)} \right) \right\} \\
 &= \Delta \left\{ \left(\frac{(2s-(a+b))}{(s-a)(s-b)} \right) - \left(\frac{c}{s(s-c)} \right) \right\} \\
 &= \Delta \left\{ \left(\frac{((a+b+c)-(a+b))}{(s-a)(s-b)} \right) - \left(\frac{c}{s(s-c)} \right) \right\} \\
 &= \Delta c \left\{ \left(\frac{1}{(s-a)(s-b)} - \frac{1}{s(s-c)} \right) \right\} \\
 &= \Delta c \left\{ \left(\frac{s(s-c) - (s-a)(s-b)}{(s-a)(s-b)s(s-c)} \right) \right\} \\
 &= \Delta c \left\{ \left(\frac{s^2 - cs - (s^2 - (a+b)s + ab)}{s(s-a)(s-b)(s-c)} \right) \right\} \\
 &= \frac{c}{\Delta} \times ((a+b-c)s - ab) \\
 &= \frac{c}{2\Delta} \times ((a+b-c)2s - 2ab) \\
 &= \frac{c}{2\Delta} \times ((a+b-c)(a+b+c) - 2ab) \\
 &= \frac{c}{2\Delta} \times ((a+b)^2 - c^2 - 2ab) \\
 &= \frac{c}{2\Delta} \times (a^2 + b^2 - c^2) \\
 &= \frac{2abc}{2\Delta} \times \left(\frac{a^2 + b^2 - c^2}{2ab} \right) \\
 &= \frac{abc}{\Delta} \times \cos(C) \\
 &= 4R \cos(C)
 \end{aligned}$$

111. Given, r_1, r_2, r_3 are in HP

$$\begin{aligned}
 \Rightarrow \quad & r_2 = \frac{2r_1r_3}{r_1+r_3} \\
 \Rightarrow \quad & \frac{\Delta}{s-b} = \frac{2 \cdot \frac{\Delta}{s-a} \cdot \frac{\Delta}{s-c}}{\frac{\Delta}{s-a} + \frac{\Delta}{s-c}} \\
 \Rightarrow \quad & \frac{1}{s-b} = \frac{2 \cdot \frac{1}{s-a} \cdot \frac{1}{s-c}}{\frac{1}{s-a} + \frac{1}{s-c}}
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \quad & \frac{1}{s-b} \left(\frac{1}{s-a} + \frac{1}{s-c} \right) = 2 \cdot \frac{1}{s-a} \cdot \frac{1}{s-c} \\
 \Rightarrow \quad & \frac{1}{s-b} \left(\frac{s-c+s-a}{(s-a)(s-c)} \right) = \frac{2}{(s-a)(s-c)} \\
 \Rightarrow \quad & (2s-a-c) = 2(s-b) \\
 \Rightarrow \quad & a+c=2b \\
 \Rightarrow \quad & a, b, c \in \text{AP}
 \end{aligned}$$

112. Since a, b, c are in AP as well as in GP, so $a = b = c$

$$\text{Now, } r_1 = \frac{\Delta}{s-a} = r_2 = r_3$$

$$\begin{aligned}
 \text{Thus, } & \left(\frac{r_1}{r_2} - \frac{r_2}{r_3} + 10 \right) \\
 &= 1 - 1 + 10 = 10
 \end{aligned}$$

113. We have

$$\begin{aligned}
 & \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} + \frac{1}{r^2} \\
 &= \frac{(s-a)^2}{\Delta^2} + \frac{(s-b)^2}{\Delta^2} + \frac{(s-c)^2}{\Delta^2} + \frac{s^2}{\Delta^2} \\
 &= \frac{1}{\Delta^2} [4s^2 - 2(a+b+c)s + (a^2 + b^2 + c^2)] \\
 &= \frac{1}{\Delta^2} [4s^2 - 2 \cdot 2s \cdot s + (a^2 + b^2 + c^2)] \\
 &= \frac{1}{\Delta^2} [4s^2 - 4s^2 + (a^2 + b^2 + c^2)] \\
 &= \frac{(a^2 + b^2 + c^2)}{\Delta^2}
 \end{aligned}$$

114. We have

$$\begin{aligned}
 & (r_1 - r)(r_2 - r)(r_3 - r) \\
 &= \left(\frac{\Delta}{(s-a)} - \frac{\Delta}{s} \right) \left(\frac{\Delta}{(s-b)} - \frac{\Delta}{s} \right) \left(\frac{\Delta}{(s-c)} - \frac{\Delta}{s} \right) \\
 &= \Delta^3 \left(\frac{1}{(s-a)} - \frac{1}{s} \right) \left(\frac{1}{(s-b)} - \frac{1}{s} \right) \left(\frac{1}{(s-c)} - \frac{1}{s} \right) \\
 &= \Delta^3 \left(\frac{s-s+a}{s(s-a)} \right) \left(\frac{s-s+b}{s(s-b)} \right) \left(\frac{s-s+c}{s(s-c)} \right) \\
 &= \frac{\Delta^3}{s^2} \left(\frac{abc}{s(s-a)(s-b)(s-c)} \right) \\
 &= \frac{\Delta^3}{s^2} \left(\frac{abc}{\Delta^2} \right) \\
 &= \frac{\Delta \times abc}{s^2} \\
 &= 4 \times \left(\frac{\Delta}{s} \right)^2 \times \left(\frac{abc}{4\Delta} \right) \\
 &= 4r^2 R
 \end{aligned}$$

115. We have

$$r_1 = \frac{\Delta}{s-a} = 1, r_2 = \frac{\Delta}{s-b} = 2$$

$$\text{and } r_3 = \frac{\Delta}{s-c}$$

$$\Rightarrow s-a = \Delta, s-b = \frac{\Delta}{2}$$

$$\text{and } s-c = \frac{\Delta}{r_3}$$

$$\Rightarrow c = \Delta \left(1 + \frac{1}{2}\right), a = \Delta \left(\frac{1}{2} + \frac{1}{r_3}\right),$$

$$b = \Delta \left(1 + \frac{1}{r_3}\right)$$

Since triangle is right angled, so

$$a^2 + b^2 = c^2$$

$$\Rightarrow \Delta^2 \left(\frac{3}{2}\right)^2 = \frac{\Delta^2 (r_3 + 2)^2}{4(r_3)^2} + \Delta^2 \left(\frac{r_3 + 1}{r_3}\right)^2$$

$$\Rightarrow \left(\frac{3}{2}\right)^2 = \frac{(r_3 + 2)^2}{4(r_3)^2} + \left(\frac{r_3 + 1}{r_3}\right)^2$$

$$\Rightarrow 9r_3^2 = (r_3 + 2)^2 + 4(r_3 + 1)^2$$

$$\Rightarrow 4r_3^2 - 12r_3 - 8 = 0$$

$$\Rightarrow r_3^2 - 3r_3 - 2 = 0$$

$$\Rightarrow r_3 = \frac{3 \pm \sqrt{17}}{2} = \frac{3 + \sqrt{17}}{2}$$

as r_3 is positive.

116. Let a, b be the sides of a triangle.

Then $a + b = 5$ and $ab = 3$

$$\text{Now, } \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\Rightarrow \cos\left(\frac{\pi}{3}\right) = \frac{19 - c^2}{6}$$

$$\Rightarrow \frac{1}{2} = \frac{19 - c^2}{6}$$

$$\Rightarrow 19 - c^2 = 3$$

$$\Rightarrow c = 4$$

$$\text{Thus, } r \cdot R = \frac{\Delta}{s} \times \frac{abc}{4\Delta}$$

$$= \frac{abc}{4s} = \frac{abc}{2(a+b+c)} = \frac{3 \cdot 4}{2(5+4)} = \frac{12}{18} = \frac{2}{3}$$

$$117. \text{ By sine rule, } \frac{a}{\sin(120^\circ)} = \frac{b}{\sin(30^\circ)}$$

$$\Rightarrow \frac{a}{\sqrt{3}/2} = \frac{b}{1/2}$$

$$\Rightarrow a = b\sqrt{3}$$

Also, from the above figure, $r = \sqrt{3}$

$$\Rightarrow \frac{\sqrt{3}}{a/2} = \tan(15^\circ)$$

$$\Rightarrow \frac{2\sqrt{3}}{a} = (2 - \sqrt{3})$$

$$\Rightarrow a = \frac{2\sqrt{3}}{(2 - \sqrt{3})}$$

$$\text{Now, } b = \frac{a}{\sqrt{3}} = \frac{2\sqrt{3}}{(2 - \sqrt{3})} \times \frac{1}{\sqrt{3}} = \frac{2}{(2 - \sqrt{3})}$$

Thus, the required area

$$= \frac{1}{2} \times ab \times \sin(30^\circ)$$

$$= \frac{1}{2} \times \frac{2\sqrt{3}}{(2 - \sqrt{3})} \times \frac{2}{(2 - \sqrt{3})} \times \frac{1}{2}$$

$$= \sqrt{3} \times (2 + \sqrt{3})^2$$

$$= \sqrt{3} \times (7 + 4\sqrt{3})$$

$$= (12 + 7\sqrt{3}) \text{ sq. u.}$$

118. Given, $r = r_1 - r_2 - r_3$

$$\Rightarrow r_1 - r = r_2 + r_3$$

$$\Rightarrow \frac{\Delta}{s-a} - \frac{\Delta}{s} = \frac{\Delta}{s-b} + \frac{\Delta}{s-c}$$

$$\Rightarrow \frac{1}{s-a} - \frac{1}{s} = \frac{1}{s-b} + \frac{1}{s-c}$$

$$\Rightarrow \frac{(s-a-s)}{(s-a)} = \frac{(s-c+s-b)}{(s-b)(s-c)}$$

$$\Rightarrow \frac{a}{(s-a)} = \frac{a}{(s-b)(s-c)}$$

$$\Rightarrow \frac{(s-b)(s-c)}{(s-a)} = 1$$

$$\Rightarrow \tan^2\left(\frac{A}{2}\right) = 1$$

$$\Rightarrow \tan\left(\frac{A}{2}\right) = 1$$

$$\Rightarrow \tan\left(\frac{A}{2}\right) = \tan\left(\frac{\pi}{4}\right)$$

$$\Rightarrow A = \frac{\pi}{2}$$

Thus, the triangle is right angled.

119. We have $r \cdot r_1 \cdot r_2 \cdot r_3$

$$= \frac{\Delta}{s} \cdot \frac{\Delta}{(s-a)} \cdot \frac{\Delta}{(s-b)} \cdot \frac{\Delta}{(s-c)}$$

$$= \frac{\Delta^4}{s(s-a)(s-b)(s-c)}$$

$$= \frac{\Delta^4}{\Delta^2}$$

$$= \Delta^2$$

120. We have

$$\begin{aligned}\frac{r_1+r_2}{1+\cos C} &= \frac{\frac{\Delta}{s-a} + \frac{\Delta}{s-b}}{2\cos^2\left(\frac{C}{2}\right)} \\ &= \frac{\frac{\Delta(s-a+s-b)}{(s-a)(s-b)}}{2\cos^2\left(\frac{C}{2}\right)} \\ &= \frac{\Delta(2s-a-b)}{(s-a)(s-b) \times 2 \times \frac{s(s-c)}{ab}} \\ &= \frac{\Delta \times abc}{s(s-a)(s-b)(s-c)} \\ &= \frac{\Delta \times abc}{\Delta^2} = \frac{abc}{\Delta}\end{aligned}$$

Similarly, we can easily prove that,

$$\frac{(r_2+r_3)}{1+\cos A} = \frac{abc}{\Delta} \text{ and } \frac{(r_3+r_1)}{1+\cos B} = \frac{abc}{\Delta}$$

$$\text{Thus, } \frac{(r_2+r_3)}{1+\cos A} = \frac{abc}{\Delta} \text{ and } \frac{(r_3+r_1)}{1+\cos B} = \frac{abc}{\Delta}$$

121. We have

$$\begin{aligned}&\left(\frac{1}{r} - \frac{1}{r_1}\right)\left(\frac{1}{r} - \frac{1}{r_2}\right)\left(\frac{1}{r} - \frac{1}{r_3}\right) \\ &= \frac{1}{\Delta^3}(s-s+a)(s-s+b)(s-s+c) \\ &= \frac{1}{\Delta^3} \times abc \\ &= 16 \times \frac{abc}{4\Delta} \times \frac{s^2}{\Delta^2(2s)^2} \\ &= \frac{16R}{r^2(a+b+c)^2}\end{aligned}$$

122. We have

$$\begin{aligned}&\left(\frac{1}{r_1} + \frac{1}{r_2}\right)\left(\frac{1}{r_2} + \frac{1}{r_3}\right)\left(\frac{1}{r_3} + \frac{1}{r_1}\right) \\ &= \left(\frac{s-a}{\Delta} + \frac{s-b}{\Delta}\right)\left(\frac{s-b}{\Delta} + \frac{s-c}{\Delta}\right)\left(\frac{s-c}{\Delta} + \frac{s-a}{\Delta}\right) \\ &= \frac{1}{\Delta^3} \times abc \\ &= \frac{1}{\left(\frac{abc}{4R}\right)^3} \times abc \\ &= \frac{64R^3}{(abc)^2}\end{aligned}$$

123. We have

$$\begin{aligned}(r_1+r_2+r_3-r)^2 &= r_1^2+r_2^2+r_3^2+r^2-2r(r_1+r_2+r_3) \\ &\quad +2(r_1r_2+r_2r_3+r_3r_1)\end{aligned}$$

$$\text{Now, } (r_1+r_2+r_3-r) = 4R$$

$$(r_1r_2+r_2r_3+r_3r_1) = s^2$$

$$\text{and } 2r(r_1+r_2+r_3)$$

$$\begin{aligned}&= 2 \times \frac{\Delta}{s} \left(\frac{\Delta}{(s-a)} + \frac{\Delta}{(s-b)} + \frac{\Delta}{(s-c)} \right) \\ &= 2 \times \frac{\Delta^2}{s} \left(\frac{1}{(s-a)} + \frac{1}{(s-b)} + \frac{1}{(s-c)} \right) \\ &= 2 \times \frac{\Delta^2}{s} \left(\frac{(s-b)(s-c) + (s-a)(s-c) + (s-a)(s-b)}{(s-a)(s-b)(s-c)} \right) \\ &= 2 \times \Delta^2 \left(\frac{3s^2 - 2(a+b+c)s + (ab+bc+ca)}{s(s-a)(s-b)(s-c)} \right) \\ &= 2 \times \Delta^2 \left(\frac{3s^2 - 2 \cdot 2s \cdot s + (ab+bc+ca)}{\Delta^2} \right)\end{aligned}$$

$$= 2 \times ((ab+bc+ca) - s^2)$$

$$\text{Thus, } (r_1^2+r_2^2+r_3^2+r^2)$$

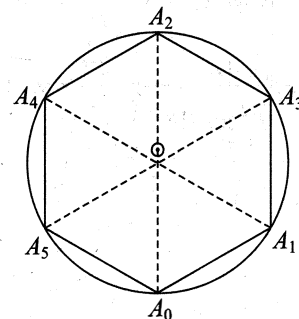
$$\begin{aligned}&= (r_1+r_2+r_3-r)^2 + 2r(r_1+r_2+r_3) \\ &\quad -2(r_1r_2+r_2r_3+r_3r_1) \\ &= 16R^2 + 2(ab+bc+ca-s^2) - 2s^2 \\ &= 16R^2 + 2(ab+bc+ca) - (2s)^2 \\ &= 16R^2 + 2(ab+bc+ca) - (a+b+c)^2 \\ &= 16R^2 - (a^2+b^2+c^2)\end{aligned}$$

Hence, the result.

 124. In a triangle ΔABC , prove that

$$\frac{(r_1+r_2)(r_2+r_3)(r_3+r_1)}{(r_1r_2+r_2r_3+r_3r_1)} = 4R$$

125.


 Here, $OA_0 = OA_1 = OA_2 = \dots = OA_5 = 1$.

and

$$\angle A_0OA_1 = \frac{2\pi}{6} = \angle A_1OA_2 = \dots = \angle A_4OA_5$$

$$\text{Now, } \cos\left(\frac{\pi}{3}\right) = \frac{OA_0^2 + OA_1^2 - A_0A_1^2}{2OA_0 \cdot OA_1}$$

$$\Rightarrow \frac{1}{2} = \frac{1+1-A_0A_1^2}{2 \cdot 1 \cdot 1}$$

$$\Rightarrow \frac{1}{2} = \frac{2 - A_0 A_1^2}{2}$$

$$\Rightarrow A_0 A_1^2 = 1$$

$$\Rightarrow A_0 A_1 = 1$$

$$\text{Also, } \cos\left(\frac{2\pi}{3}\right) = \frac{OA_0^2 + OA_1^2 - A_0 A_2^2}{2 \cdot OA_0 \cdot OA_1}$$

$$\Rightarrow -\frac{1}{2} = \frac{1 + 1 - A_0 A_2^2}{2 \cdot 1 \cdot 1}$$

$$\Rightarrow \frac{1}{2} = \frac{2 - A_0 A_2^2}{2}$$

$$\Rightarrow A_0 A_2 = \sqrt{3}$$

$$\text{Again, } \cos\left(\frac{4\pi}{3}\right) = \frac{OA_0^2 + OA_4^2 - A_0 A_4^2}{2 \cdot OA_0 \cdot OA_4}$$

$$= \frac{1 + 1 - A_0 A_4^2}{2 \cdot 1 \cdot 1}$$

$$= \frac{2 - A_0 A_4^2}{2}$$

$$\Rightarrow -\frac{1}{2} = \frac{2 - A_0 A_4^2}{2}$$

$$\Rightarrow A_0 A_4 = \sqrt{3}$$

Hence, the value of

$$\Rightarrow A_0 A_1 \cdot A_0 A_2 \cdot A_0 A_4 = 1 \cdot \sqrt{3} \cdot \sqrt{3} = 3$$

126. Let R be the radius of the circle.

$$\text{Then, } A_1 = \pi R^2$$

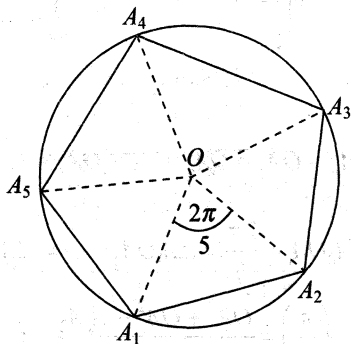
$$\text{and } \Delta = \frac{nR^2}{2} \sin\left(\frac{2\pi}{n}\right)$$

$$A_2 = \frac{5 \cdot R^2}{2} \sin\left(\frac{360^\circ}{5}\right)$$

$$= \frac{5}{2} R^2 \sin(72^\circ) = \frac{5R^2}{2} \times \cos(18^\circ)$$

$$\text{Now, } \frac{A_1}{A_2} = \frac{\pi R^2}{\frac{5}{2} R^2 \cos(18^\circ)} = \frac{2\pi}{5} \times \sec(18^\circ)$$

127.



$$\text{Here, } OA_1 = OA_2 = OA_3 = OA_4 = OA_5 = 1$$

and

$$\angle A_1 O A_2 = \frac{2\pi}{5} = \angle A_2 O A_3 = \dots = \angle A_4 O A_5$$

$$\text{Now, } \cos\left(\frac{2\pi}{5}\right) = \frac{OA_1^2 + OA_2^2 - A_1 A_2^2}{2 \cdot OA_1 \cdot OA_2}$$

$$\Rightarrow \sin(18^\circ) = \frac{1 + 1 - A_1 A_2^2}{2 \cdot 1 \cdot 1}$$

$$\Rightarrow \frac{\sqrt{5} - 1}{4} = \frac{2 - A_1 A_2^2}{2}$$

$$\Rightarrow A_1 A_2^2 = 2 - \frac{\sqrt{5} - 1}{2} = \frac{5 - \sqrt{5}}{2}$$

$$\Rightarrow A_1 A_2 = \sqrt{\frac{5 - \sqrt{5}}{2}}$$

$$\text{Similarly, } A_1 A_3 = \sqrt{\frac{5 + \sqrt{5}}{2}}$$

$$\text{Thus, } A_1 A_2 \times A_1 A_3$$

$$= \sqrt{\left(\frac{5 - \sqrt{5}}{2}\right) \times \left(\frac{5 + \sqrt{5}}{2}\right)} = \sqrt{\frac{25 - 5}{4}} = \sqrt{\frac{20}{4}} = \sqrt{5}$$

$$128. R = \frac{a}{2} \operatorname{cosec}\left(\frac{\pi}{n}\right),$$

As we know that, the circum-radius of n sided regular polygon

$$= \frac{a}{2} \operatorname{cosec}\left(\frac{\pi}{n}\right), \text{ where } a = \text{side}$$

and n = number of sides

$$= 6 \times \operatorname{cosec}\left(\frac{\pi}{12}\right)$$

$$= 6 \times \operatorname{cosec}(15^\circ)$$

$$= \frac{6 \times 2\sqrt{2}}{\sqrt{3} - 1}$$

$$= 6\sqrt{2}(\sqrt{3} + 1)$$

129. Let the perimeter of the pentagon and the decagon be $10x$.

Then each side of the pentagon is $2x$ and the decagon is x .

Let A_1 = the area of the pentagon

$$= 5x^2 \cot\left(\frac{\pi}{5}\right)$$

and A_2 = the area of the decagon

$$= \frac{5}{2} x^2 \cot\left(\frac{\pi}{10}\right)$$

$$\text{Now, } \frac{A_1}{A_2} = \frac{5x^2 \cot\left(\frac{\pi}{5}\right)}{\frac{5}{2} x^2 \cot\left(\frac{\pi}{10}\right)} = \frac{2 \cot\left(\frac{\pi}{5}\right)}{\cot\left(\frac{\pi}{10}\right)}$$

$$\begin{aligned}
 &= \frac{2 \cot(36^\circ)}{\cot(18^\circ)} = \frac{2 \cos(36^\circ) \sin(18^\circ)}{\sin(36^\circ) \cos(18^\circ)} \\
 &= \frac{2 \cos(36^\circ) \sin(18^\circ)}{2 \sin(18^\circ) \cos^2(18^\circ)} \\
 &= \frac{2 \cos(36^\circ)}{(1 + \cos(36^\circ))} \\
 &= \frac{2 \left(\frac{\sqrt{5}+1}{4} \right)}{\left(1 + \frac{\sqrt{5}+1}{4} \right)} \\
 &= \frac{2(\sqrt{5}+1)}{(\sqrt{5}+5)} \\
 &= \frac{2(\sqrt{5}+1)}{\sqrt{5}(\sqrt{5}+1)} \\
 &= \frac{2}{\sqrt{5}}
 \end{aligned}$$

130. We have

$$R = \frac{2a}{2} \operatorname{cosec}\left(\frac{\pi}{n}\right) = a \operatorname{cosec}\left(\frac{\pi}{n}\right)$$

$$\text{and } r = \frac{2a}{2} \cot\left(\frac{\pi}{n}\right) = a \cot\left(\frac{\pi}{n}\right)$$

Now,

$$\begin{aligned}
 r + R &= a \cot\left(\frac{\pi}{n}\right) + a \operatorname{cosec}\left(\frac{\pi}{n}\right) \\
 &= a \left(\cot\left(\frac{\pi}{n}\right) + \operatorname{cosec}\left(\frac{\pi}{n}\right) \right) \\
 &= a \left(\frac{1 + \cos\left(\frac{\pi}{n}\right)}{\sin\left(\frac{\pi}{n}\right)} \right) \\
 &= a \left(\frac{2 \cos^2\left(\frac{\pi}{2n}\right)}{2 \sin\left(\frac{\pi}{2n}\right) \cos\left(\frac{\pi}{2n}\right)} \right) \\
 &= a \cot\left(\frac{\pi}{2n}\right)
 \end{aligned}$$

131. Let A_1 be the area of the regular pentagon and A_2 be the area of the regular decagon.

Therefore, $A_1 = A_2$

$$\Rightarrow \frac{5a^2}{4} \cot\left(\frac{\pi}{5}\right) = \frac{6b^2}{4} \cot\left(\frac{\pi}{6}\right)$$

$$\Rightarrow 5a^2 \cot\left(\frac{\pi}{5}\right) = 6b^2 \cot\left(\frac{\pi}{6}\right)$$

$$\Rightarrow 5a^2 \cot(36^\circ) = 6b^2 \cot(30^\circ)$$

$$\Rightarrow 5a^2 \cot(36^\circ) = 6\sqrt{3}b^2$$

$$\Rightarrow \frac{a^2}{6\sqrt{3}} = \frac{b^2}{5 \cot(36^\circ)} = \lambda$$

Hence the ratio of their perimeters

$$\begin{aligned}
 &= \frac{5a}{6b} \\
 &= \frac{5}{6} \times \frac{\sqrt{\lambda 6\sqrt{3}}}{\sqrt{5\lambda \cot(36^\circ)}} = \sqrt{\frac{5\sqrt{3}}{6} \tan(36^\circ)}
 \end{aligned}$$

132. Let the perimeter of the two polygons are nx and $2nx$ respectively.

Then each side of the polygons are $2x$ and x .

Let A_1 = the area of the polygon of n sides

$$= nx^2 \cot\left(\frac{\pi}{n}\right)$$

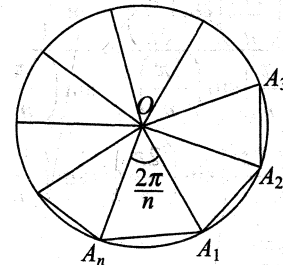
and A_2 = the area of the decagon

$$= \frac{5}{2} n^2 \cot\left(\frac{\pi}{2n}\right)$$

$$\text{Thus, } \frac{A_1}{A_2} = \frac{5n^2 \cot\left(\frac{\pi}{n}\right)}{\frac{5}{2} n^2 \cot\left(\frac{\pi}{2n}\right)} = \frac{2 \cot\left(\frac{\pi}{n}\right)}{\cot\left(\frac{\pi}{2n}\right)}$$

$$\begin{aligned}
 &= \frac{2 \cos\left(\frac{\pi}{n}\right) \sin\left(\frac{\pi}{2n}\right)}{2 \sin\left(\frac{\pi}{2n}\right) \cos^2\left(\frac{\pi}{2n}\right)} \\
 &= \frac{2 \cos\left(\frac{\pi}{n}\right)}{1 + \cos\left(\frac{\pi}{n}\right)}
 \end{aligned}$$

133. Let O be the centre and $A_1 A_2 \dots A_n$ be the regular polygon of n -sides.



Let $OA_1 = OA_2 = \dots = OA_n = r$

and $\angle A_1 O A_2 = \angle A_2 O A_3$

$$= \dots = \angle A_n O A_1 = \frac{2\pi}{n}$$

From the triangle $OA_1 A_2$,

$$\begin{aligned}
 \cos\left(\frac{2\pi}{n}\right) &= \frac{OA_1^2 + OA_2^2 - A_1 A_2^2}{2 \cdot OA_1 \cdot OA_2} \\
 &= \frac{r^2 + r^2 - A_1 A_2^2}{2 \cdot r \cdot r}
 \end{aligned}$$

$$\Rightarrow A_1 A_2^2 = 2r^2 - 2r^2 \cos\left(\frac{2\pi}{n}\right)$$

$$\Rightarrow A_1 A_2^2 = 2r^2 \left(1 - \cos\left(\frac{2\pi}{n}\right)\right)$$

$$\Rightarrow A_1 A_2^2 = 2r^2 \cdot 2 \sin^2\left(\frac{2\pi}{n}\right) \\ = 4r^2 \cdot \sin^2\left(\frac{2\pi}{n}\right)$$

$$\Rightarrow A_1 A_2 = 2r \cdot \sin\left(\frac{2\pi}{n}\right)$$

$$\text{Similarly, } A_1 A_3 = 2r \cdot \sin\left(\frac{4\pi}{n}\right)$$

$$\text{and } A_1 A_4 = 2r \cdot \sin\left(\frac{6\pi}{n}\right)$$

$$\text{Given, } \frac{1}{A_1 A_2} = \frac{1}{A_1 A_3} + \frac{1}{A_1 A_4}$$

$$\Rightarrow \frac{1}{2r \cdot \sin\left(\frac{2\pi}{n}\right)} = \frac{1}{2r \cdot \sin\left(\frac{4\pi}{n}\right)} + \frac{1}{2r \cdot \sin\left(\frac{6\pi}{n}\right)}$$

$$\Rightarrow \frac{1}{\sin\left(\frac{\pi}{n}\right)} - \frac{1}{\sin\left(\frac{2\pi}{n}\right)} + \frac{1}{\sin\left(\frac{3\pi}{n}\right)}$$

$$\Rightarrow \frac{1}{\sin\left(\frac{\pi}{n}\right)} - \frac{1}{\sin\left(\frac{3\pi}{n}\right)} = \frac{1}{\sin\left(\frac{2\pi}{n}\right)}$$

$$\Rightarrow \frac{\sin\left(\frac{3\pi}{n}\right) - \sin\left(\frac{\pi}{n}\right)}{\sin\left(\frac{\pi}{n}\right) \sin\left(\frac{3\pi}{n}\right)} = \frac{1}{\sin\left(\frac{2\pi}{n}\right)}$$

$$\Rightarrow \frac{2 \cos\left(\frac{2\pi}{n}\right) \sin\left(\frac{\pi}{n}\right)}{\sin\left(\frac{\pi}{n}\right) \sin\left(\frac{3\pi}{n}\right)} = \frac{1}{\sin\left(\frac{2\pi}{n}\right)}$$

$$\Rightarrow 2 \cos\left(\frac{2\pi}{n}\right) \sin\left(\frac{2\pi}{n}\right) = \sin\left(\frac{3\pi}{n}\right)$$

$$\Rightarrow \sin\left(\frac{4\pi}{n}\right) = \sin\left(\frac{3\pi}{n}\right)$$

$$\Rightarrow \sin\left(\frac{4\pi}{n}\right) = \sin\left(\pi - \frac{3\pi}{n}\right)$$

$$\Rightarrow \left(\frac{4\pi}{n}\right) = \left(\pi - \frac{3\pi}{n}\right)$$

$$\Rightarrow \left(\frac{7\pi}{n}\right) = \pi$$

$$\Rightarrow n = 7$$

134. Let r be the radius of the in-circle and r_1, r_2 and r_3 are the ex-radii of the given triangle.

$$\text{Then } \frac{1}{\sqrt{A_1}} + \frac{1}{\sqrt{A_2}} + \frac{1}{\sqrt{A_3}} \\ = \frac{1}{\sqrt{\pi r_1^2}} + \frac{1}{\sqrt{\pi r_2^2}} + \frac{1}{\sqrt{\pi r_3^2}} \\ = \frac{1}{\sqrt{\pi}} \left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \right) \\ = \frac{1}{\sqrt{\pi}} \times \frac{1}{r} \\ = \frac{1}{\sqrt{\pi r^2}} \\ = \frac{1}{\sqrt{A}}$$

Hence, the result.

135. Let the perimeter of the polygon of n sides = nx

Let A_1 = the area of the polygon of n sides

$$= nx^2 \cot\left(\frac{\pi}{n}\right)$$

and A_2 = the area of the circle = πx^2

$$\text{Now, } \frac{A_2}{A_1} = \frac{\pi x^2}{nx^2 \cot\left(\frac{\pi}{n}\right)} = \frac{\tan\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}}$$

$$\text{Thus, } A_2 : A_1 = \tan\left(\frac{\pi}{n}\right) : \left(\frac{\pi}{n}\right)$$

136. We have

$$r + R = \frac{a}{2} \cot\left(\frac{a}{n}\right) + \frac{a}{2} \operatorname{cosec}\left(\frac{a}{n}\right) \\ = \frac{a}{2} \left[\cot\left(\frac{a}{n}\right) + \operatorname{cosec}\left(\frac{a}{n}\right) \right] \\ = \frac{a}{2} \left[\frac{\cos\left(\frac{a}{n}\right)}{\sin\left(\frac{a}{n}\right)} + \frac{1}{\sin\left(\frac{a}{n}\right)} \right] \\ = \frac{a}{2} \left[\frac{1 + \cos\left(\frac{a}{n}\right)}{\sin\left(\frac{a}{n}\right)} \right] \\ = \frac{a}{2} \left[\frac{2 \cos^2\left(\frac{a}{2n}\right)}{2 \sin\left(\frac{a}{2n}\right) \cos\left(\frac{a}{2n}\right)} \right] \\ = \frac{a}{2} \cot\left(\frac{a}{2n}\right)$$